

Chapter 0. 4章 { (1) 波动/双曲. 热/抛物. Laplace/椭圆 2阶线性 PDE.
 ①. $u_{tt} = a^2 u_{xx} + f(x,t)$
 ②. $u_t = a^2 u_{xx} + f(x,t)$
 ③. $\Delta u = u_{xx} + u_{yy} = 0$
 (1). PDE 分类.

ODE. $u(x)$. PDE: 多个未知量. $u(x,y), u_x, u_y, u_{xx}, u_{yy}, u_{xy}, \dots$.

阶数: 微分最高次.

线性 L.I. $\Leftrightarrow u$ 关于每个微分都线性. 线性 $\Rightarrow$ 叠加性.

非齐次: 自由项 $f(x,y,t) \neq 0$.

发展/动态方程: $u(x,t) \rightarrow$ 稳, t 不显性
 泛定/静态方程: $u(x,y)$.

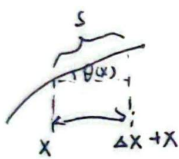
波动, 热 $\rightarrow t$ 依赖. 发展.
 Laplace $\rightarrow t$ 不依赖. 静态.

Chapter 1.

1. 弦振动方程 $\rightarrow$ 波动方程.

弦做微扰横振动.

Δx 振动方向竖直, 水平受力 $= 0$.



$u(x,t)$: 横位移.

$s(x,t)$: 弦长.

$T(x,t)$: 弦张力.

$F(x,t)$: 外力密度方程.

$\Rightarrow$ 由动量定理 $m \cdot \Delta V = \hat{F} \cdot \Delta t$.

$$\rho \cdot \Delta x \cdot [u_t(t+\Delta t) - u_t(t)] = \Delta t [T[u_x(x+\Delta x) - u_x(x)] + F \cdot \Delta x]$$

$$u_{tt} = \frac{T}{\rho} u_{xx} + \frac{F}{\rho}$$

$$u_{tt} = \underbrace{a^2}_{>0} u_{xx} + f(x,t)$$

波动方程.

$$\textcircled{1} m = \rho \cdot \Delta s = \rho \cdot \int_x^{x+\Delta x} \sqrt{1+(u_x)^2} dx \approx \rho \cdot \Delta x \cdot (|u_x| = \tan \theta < 1)$$

$$\textcircled{2} \text{速度变化量 } \Delta V = u_t(t+\Delta t) - u_t(t)$$

$$\textcircled{3} T(x) \cdot \cos(\theta) = T(x+\Delta x) \cdot \cos(\theta+\Delta\theta)$$

$$\cos(\theta) \cdot \cos(\theta+\Delta\theta) \rightarrow 1 \Rightarrow T(x) \approx T(x+\Delta x)$$

i.e. 理想模型下张力与位置无关.

$$\textcircled{4} \text{垂直方向合力 } \hat{F} = T[u_x(x+\Delta x) - u_x(x)] + F \cdot \Delta x$$

2. 性质.

①. IC. 只有 IC $\Rightarrow$ Cauchy / 自由振动 / 全空间问题.

Theo. u 对 t 有 k 阶导 $\Rightarrow$ 需要 k 个 IC.

$$\text{e.g. } u_{tt} = \dots \Rightarrow \begin{cases} u(x,0) = p_1(x) \\ u_t(x,0) = p_2(x) \end{cases} \quad \text{IC} = \frac{\partial^i u}{\partial t^i} \Big|_{t=0}, i=0, \dots, k-1$$

②. BC.

第一类 / Dirichlet $u|_{x=l} = f_1(t)$.

第二类 / Neumann $u_x|_{x=l} = f_2(t)$

第三类 / Robin $(u+u_x)|_{x=l} = f_1(t) + f_2(t)$

③. 定解问题的解的存在性: 存在, 唯一, 稳定.

④. 叠加原理.

$$\left. \begin{aligned} \text{(i)} \quad u &= \sum_{k=1}^n u_k \\ u_{tt} &= a^2 u_{xx} + f(x, t) \end{aligned} \right\} \Rightarrow u_{tt} = \sum_{k=1}^n u_{k,tt} = a^2 (\sum u_k)_{xx} + f(x, t)$$

定义线性算子 $L: L[u] = u_{tt} - a^2 u_{xx} = \begin{cases} 0 & \text{齐次} \\ f(x, t) & \text{非齐次} \end{cases}$

齐次时, $u = \sum c_i u_i$

$\Rightarrow L[u] = L[u_i] = 0 \Rightarrow u$ 也是解.

非齐次时, u_i 是解 i.e. $u_{i,tt} = a^2 u_{i,xx} + f_i$, $u = \sum u_i$

$$\Rightarrow u_{tt} = \sum u_{i,tt} = a^2 \sum u_{i,xx} + \sum f_i = a^2 u_{xx} + \sum f_i$$

$$\Rightarrow L[u] = L[u_i]$$

独立扰动叠加.

(ii). 级数在求解范围内收敛则叠加原理可推广至无穷.

$$u = \sum_{k=1}^{\infty} c_k u_k(x, t).$$

(iii) $u(x, t, M)$ 有 $u_t = a^2 u_{xx} + f(x, t, M)$, $u(x, t) = \int_D u(x, t, M) dM$.

$$\Rightarrow u_{tt} = a^2 u_{xx} + \int_D f(x, t, M) dM.$$

(iv). 叠加原理方便拆分.

$$u(x, t) = v(x, t) + w(x, t)$$

$$\begin{cases} u_t = v_t + w_t = a^2 (v_{xx} + w_{xx}) + f = a^2 (v+w)_{xx} + f = a^2 u_{xx} + f, \\ \text{IC: } u(0, x) = \varphi(x) + 0 = v(0, x) + w(0, x) \\ \text{BC: } u|_{x=0} = v|_{x=0} + w|_{x=0}, \quad u_x|_{x=l} = v_x|_{x=l} + w_x|_{x=l} \end{cases}$$

3. 解线性无限弦长波动 E / 柯西问题.

①. 初值条件的非齐次性 $\rightarrow$ 叠加原理分离问题.

$$\text{Cauchy} \Rightarrow \text{无 BC. 原问题: } \begin{cases} u_{tt} = a^2 u_{xx} + f(x, t), (x, t) \in \mathbb{R} \times \mathbb{R}^+ \\ \text{IC: } u|_{t=0} = \varphi(x), u_t|_{t=0} = \psi(x). \end{cases}$$

$$\text{叠加原理分离} \Rightarrow \begin{cases} \text{① N-H IC \& H-E} \rightarrow \text{D'Alembert 公式.} \\ \text{② H-IC \& N-H-E} \end{cases}$$

②. D'Alembert 公式 $\rightarrow$ 解 N-H-IC, H-E 柯西问题.

$$v(x, t) = \frac{1}{2} [\varphi(x+at) + \varphi(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\xi) d\xi.$$

性质. (i). 解存在.

① 存在: 公式本身.

② 唯一: 设 $u_1 - u_2 = u \neq 0$. $\Rightarrow \begin{cases} u_t = a^2 u_{xx}, x \in \mathbb{R}, \\ u(x, 0) = u_1(x, 0) - u_2(x, 0) \\ u_t(x, 0) = u_{1t}(x, 0) - u_{2t}(x, 0) \end{cases} \Rightarrow \text{物理意义.}$

③ 稳定: 设 $\begin{cases} IC_1: \varphi_1, \psi_1 \Rightarrow u_1 \\ IC_2: \varphi_2, \psi_2 \Rightarrow u_2 \end{cases}$

$$\Rightarrow |u_1 - u_2| = \frac{1}{2} \left\{ [\varphi_1(x+at) + \varphi_1(x-at)] - [\varphi_2(x+at) + \varphi_2(x-at)] \right\} + \frac{1}{2a} \int_{x-at}^{x+at} |\psi_1 - \psi_2| dz.$$

设 $|\varphi_1 - \varphi_2| < \delta$, $|\psi_1 - \psi_2| < \delta$, 则

$$\forall \varepsilon > 0, \exists \delta = \frac{\varepsilon}{1+\gamma}, \forall |\varphi_1 - \varphi_2| < \delta, |\psi_1 - \psi_2| < \delta,$$

$$\text{有 } |u_1 - u_2| \leq \frac{1}{2} \delta + \frac{1}{2} \delta + \frac{1}{2a} \cdot \delta \cdot 2a \cdot \gamma < \delta(1+\gamma) = \varepsilon.$$

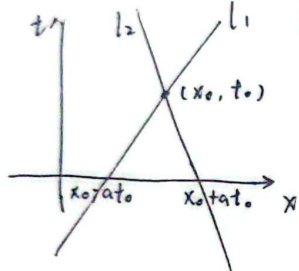
此处讨论有限时间内的稳定性. 即非 AS.

(ii). $x+at$, $x-at$ 物理解释.

↑
左行波, 右行波.

波动方程由行波叠加得到. D'Alembert 的解是行波解.

特征线 C.L.: $\begin{cases} x-at = C_1 \\ x+at = C_2 \end{cases}$

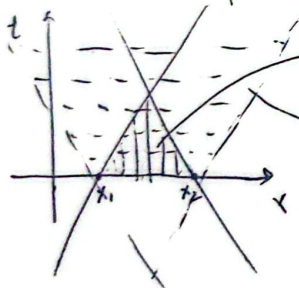


$x+at: C_2$

$$l_1: x-at = x_0 - at_0$$

$$l_2: x+at = x_0 + at_0$$

$\Rightarrow$ 给定任一点, 波动 E 由 IC 在该点的 依赖区间 $[x_0 - at_0, x_0 + at_0]$ 决定.



阴影内任一点, f 由 IC 在 $[x_1, x_2]$ 决定, 阴影为 决定域.

外侧 C.L. 形成梯形是 影响域. 初值 E 在影响域受 $[x_1, x_2]$ 上扰动的影啊.

③. Duhamel 原理 $\rightarrow$ H-IC. N-H.E 柯西.

$$w(x, t) = \int_0^t h(x, t, z) dz = \frac{1}{2a} \int_0^t \int_{x-a(t-z)}^{x+a(t-z)} f(z, \tau) dz d\tau.$$

④. Kirchhoff 公式 $\rightarrow$ N-H-IC. N-H-Cauchy.

$$\text{对于 } \begin{cases} u_{tt} - a^2 u_{xx} = f(x, t), (x, t) \in \mathbb{R} \times \mathbb{R}^+ \\ u|_{t=0} = \varphi(x), u_t|_{t=0} = \psi(x) \end{cases}$$

$$\text{有 } \boxed{u(x, t) = \frac{1}{2} (\varphi(x-at) + \varphi(x+at)) + \frac{1}{2a} \int_{x-at}^{x+at} \psi(\alpha) d\alpha + \frac{1}{2a} \int_0^t \int_{x-a(t-\tau)}^{x+a(t-\tau)} f(\xi, \tau) d\xi d\tau.}$$

4. 解线性初值问题波动方程.

$$\text{1维. } \begin{cases} u_{tt} = a^2 u_{xx} + f(x, t), x \in [0, l], t > 0 \\ \text{IC: } u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x) \\ \text{BC: } u(0, t) = p(t), u_t(l, t) = q(t) \end{cases}$$

(D) (N)

① 3类 BC 可任意组合 (结合具体物理背景)

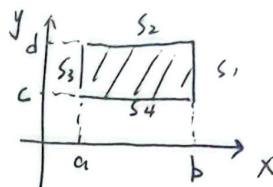
② 从造定角度, 边界点, 含 BC 即可.

③ BC 与 IC 需耦合.

$$\text{矛盾例子: } \begin{cases} u(x, 0) = d \\ u(l, t) = c, c \neq d \end{cases}$$

此时 PDE 不可能连续.

$$\text{二维. } \begin{cases} u_{tt} = a^2 \Delta u, (x, y) \in (a, b) \times (c, d) \\ \text{IC: } u(x, y, 0) = \varphi(x, y) \\ u_t(x, y, 0) = \psi(x, y) \\ \text{BC: } S_1: u(b, y, t) = p_1(y, t), y \in (c, d), (D) \\ S_2: u_y(x, d, t) = p_2(x, t), (N) \end{cases}$$



IC, BC 在特殊点 注意耦合.

①. 半无限弦长: 端点 = 0, $x \in (0, +\infty)$.

解半无限弦长问题: BC 齐次化 + 问题延拓.

N-H-BC 半无限弦长 $\xrightarrow{\text{BC 齐次化}}$ H-BC 无限弦长.

$$\begin{cases} u_{tt} = a^2 u_{xx} + f(x, t) \\ \text{IC: } u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x). \\ \text{BC: } u(0, t) = p(t). \end{cases}$$

对线性 Cauchy 有.

$$\text{取 } v(x, t) = u(x, t) - p(t).$$

$$\Rightarrow \begin{cases} u_{tt} + p''(t) = a^2 v_{xx} + [f(x, t) - p''(t)], x \in \mathbb{R} \\ v_t(x, 0) = u_t(x, 0) - p'(0) = \psi(x) - p'(0) \\ v(0, t) = 0. \end{cases}$$

(i). 奇偶性. (ii). f, φ, ψ 奇 $\rightarrow u$ 奇.

(ii). f, φ, ψ 偶 $\rightarrow u$ 偶.

奇偶性解 H-BC 半无限弦长: 对于半无限弦长问题, 作延拓建立无限弦长问题, 对于原问题的解答为取一半.

(i) H-BC (D) $\rightarrow$ 问题作奇延拓.

(ii) H-BC (N) $\rightarrow$ 问题作偶延拓.

②. 有限弦长. 端点 ~~固定~~ 0, 1. $x \in (0, 1)$

N-H-BC 有限弦长 $\xrightarrow{\text{BC 齐次化.}}$ H-BC 有限弦长.

$$\begin{cases} u_{tt} = a^2 u_{xx} + f(x, t) \\ \text{IC: } u(x, 0) = \varphi(x), u_t(x, 0) = \psi(x) \\ \text{BC: } u(0, t) = u_1(t), u(1, t) = u_2(t) \end{cases}$$

2 点插值构造 $h(x, t) = \frac{1-x}{l} u_1(t) + \frac{x}{l} u_2(t)$

$$\text{s.t. } \begin{cases} h(0, t) = u_1(t) \\ h(1, t) = u_2(t) \end{cases}$$

$$\Rightarrow z(x, t) = u(x, t) - h(x, t) \text{ 有 } \begin{cases} z(0, t) = 0, \\ z(1, t) = 0. \end{cases}$$

$$\Rightarrow \text{分解为 } \begin{cases} v_{tt} = a^2 v_{xx} \\ v(x, 0) = \varphi(x), v_t(x, 0) = \psi(x) \\ v(0, t) = 0, v(1, t) = 0. \end{cases}$$

H-BC. N-H-IC. H-E.

$$\begin{cases} w_{tt} = a^2 w_{xx} + f(x, t) \\ w(x, 0) = 0, w_t(x, 0) = 0, \\ w(0, t) = 0, w(1, t) = 0. \end{cases}$$

H-BC. H-IC. N-H-E

(1). 分离变量法 $\rightarrow$ H-BC. N-H-IC. H-E.

$$\begin{cases} v_{tt} = a^2 v_{xx}, x \in (0, 1) \\ v(x, 0) = \varphi(x), v_t(x, 0) = \psi(x) \\ v(0, t) = 0, v(1, t) = 0. \end{cases}$$

[解]. 设 $V(x, t)$ 能分离成 2 个单变量函数 i.e. $v(x, t) = X(x)T(t)$. $X, T \neq 0$.

$$\Rightarrow X \cdot T'' = a^2 X'' \cdot T$$

$$\Rightarrow \frac{T''}{a^2 T} = \frac{X''}{X} = -\lambda, \text{ 有}$$

$$\begin{cases} X'' + \lambda X = 0, & \text{BC: } v(0, t) = X(0) \cdot T(t) = 0 \Rightarrow X(0) = 0, \\ T'' + a^2 \lambda T = 0, & v(1, t) = X(1) \cdot T(t) = 0 \Rightarrow X(1) = 0. \end{cases}$$

$$\Rightarrow \begin{cases} X'' + \lambda X = 0 \\ X(0) = 0, X(1) = 0. \end{cases}$$

①. $\lambda = 0$. $X'' = 0 \Rightarrow X = Ax + B$.

$$X(0) = 0 \Rightarrow B = 0, X(1) = 0 \Rightarrow A = 0 \Rightarrow X \equiv 0.$$

②. $\lambda < 0$. 解特征方程 $\Rightarrow X = A \cdot e^{-\sqrt{\lambda}x} + B \cdot e^{\sqrt{\lambda}x}$.

$$\begin{aligned} X(0) = A + B = 0, X(1) = A \cdot e^{-\sqrt{\lambda}} + B \cdot e^{\sqrt{\lambda}} = 0 \\ \Rightarrow A \cdot (e^{\sqrt{\lambda}} - e^{-\sqrt{\lambda}}) = 0 \Rightarrow A = 0, B = 0 \Rightarrow X \equiv 0. \end{aligned}$$

③. $\lambda > 0$. 解特征方程 $\Rightarrow X = (C_1 + C_2) \cos(\sqrt{\lambda}x) + i \cdot (C_1 - C_2) \sin(\sqrt{\lambda}x)$

$$= A \cos(\sqrt{\lambda}x) + B \sin(\sqrt{\lambda}x)$$

$$X(0) = A = 0, X(1) = B \sin(\sqrt{\lambda}) = 0 \Rightarrow \lambda = \left(\frac{k\pi}{l}\right)^2, k = 1, \dots$$

$$\Rightarrow \lambda_k = \left(\frac{k\pi}{l}\right)^2, k = 1, \dots$$

$$\Rightarrow X_k = A_k \sin\left(\frac{k\pi}{l}x\right), k = 1, \dots$$

特征值

特征函数.

$$\lambda > 0. T_k'' + a^2 \lambda_k T_k = 0 \Rightarrow T_k = B_k \cos(a \sqrt{\lambda_k} t) + C_k \sin(a \sqrt{\lambda_k} t) \\ = B_k \cdot \cos\left(\frac{a k \pi}{l} t\right) + C_k \sin\left(\frac{a k \pi}{l} t\right)$$

$$\Rightarrow V_k = X_k T_k = A_k \sin\left(\frac{k \pi}{l} x\right) \left[B_k \cdot \cos\left(\frac{a k \pi}{l} t\right) + C_k \cdot \sin\left(\frac{a k \pi}{l} t\right) \right] \\ = \sin\left(\frac{k \pi}{l} x\right) \left[A_k \cdot \cos\left(\frac{a k \pi}{l} t\right) + B_k \cdot \sin\left(\frac{a k \pi}{l} t\right) \right], k=1, \dots$$

$$\Rightarrow V = \sum_{k=1}^{\infty} V_k = \sum_{k=1}^{\infty} \sin\left(\frac{k \pi}{l} x\right) \left[A_k \cdot \cos\left(\frac{a k \pi}{l} t\right) + B_k \cdot \sin\left(\frac{a k \pi}{l} t\right) \right], k=1, \dots$$

$$2C. V(x, 0) = \sum_{k=1}^{\infty} A_k \left(\sin \frac{k \pi x}{l} \right) = \varphi(x).$$

$$A_k \int_0^l \left(\sin \frac{k \pi x}{l} \right) \left(\sin \frac{k \pi x}{l} \right) dx = \int_0^l \varphi(x) \left(\sin \frac{k \pi x}{l} \right) dx = A_k \cdot \frac{l}{2} \delta_{nk}.$$

$$\Rightarrow A_k = \frac{2}{l} \int_0^l \varphi(x) \cdot \left(\sin \frac{k \pi x}{l} \right) dx$$

$$V_t = \sum_{k=1}^{\infty} \left(\cos \frac{k \pi x}{l} \right) \cdot \left(A_k \cdot \frac{k \pi a}{l} \right) \cdot \left(\sin \frac{k \pi a}{l} t \right) + \left(\sin \frac{k \pi x}{l} \right) \cdot \left(B_k \cdot \frac{k \pi a}{l} \right) \cdot \left(\cos \frac{k \pi a}{l} t \right)$$

$$V_t(x, 0) = \sum_{k=1}^{\infty} \left(B_k \cdot \frac{k \pi a}{l} \right) \left(\sin \frac{k \pi x}{l} \right) = \psi(x).$$

$$B_k \cdot \frac{k \pi a}{l} \int_0^l \left(\sin \frac{k \pi x}{l} \right) \left(\sin \frac{k \pi x}{l} \right) dx = \int_0^l \psi(x) \left(\sin \frac{k \pi x}{l} \right) dx$$

$$B_k \cdot \frac{k \pi a}{l} \cdot \frac{l}{2} \cdot \delta_{nk} = \int_0^l \psi(x) \left(\sin \frac{k \pi x}{l} \right) dx$$

$$\Rightarrow B_k = \frac{k \pi a}{2} \int_0^l \psi(x) \cdot \sin\left(\frac{k \pi x}{l}\right) dx.$$

$$\Rightarrow V = \sum_{k=1}^{\infty} \sin\left(\frac{k \pi}{l} x\right) \left[A_k \cdot \cos\left(\frac{a k \pi}{l} t\right) + B_k \cdot \sin\left(\frac{a k \pi}{l} t\right) \right]$$

$$= \sum_{k=1}^{\infty} \sin\left(\frac{k \pi}{l} x\right) \left[\frac{2}{l} \cos\left(\frac{a k \pi}{l} t\right) \left(\int_0^l \varphi(x) \cdot \left(\sin \frac{k \pi x}{l} \right) dx \right) \right.$$

$$\left. + \frac{k \pi a}{2} \sin\left(\frac{a k \pi}{l} t\right) \left(\int_0^l \psi(x) \cdot \sin\left(\frac{k \pi x}{l}\right) dx \right) \right], k=1, \dots$$

□

Chapter 3.

泊松 $\begin{cases} 2D: \Delta u = u_{xx} + u_{yy} = f(x, y) \\ 3D: \Delta u = u_{xx} + u_{yy} + u_{zz} = f(x, y, z) \end{cases}$

1. 分离变量法 $\rightarrow$ N-H-B.C. Lap.

(p2) ①. N-H-B.C. Lap.
(18 (N))

$$\begin{cases} u_{yy} + u_{xx} = 0, & x \in (0, a), y \in (0, b) \\ u(x, 0) = f(x), u(x, b) = g(x) \\ u_x(0, y) = u_x(a, y) = 0 \end{cases}$$

[解]. $u = XY, XY'' + X''Y = 0.$

$$XY'' = -X''Y, \quad \frac{X''}{X} = -\frac{Y''}{Y} = -\lambda.$$

$$\begin{cases} X'' + \lambda X = 0 \\ Y'' - \lambda Y = 0 \end{cases} \quad \begin{cases} X'(0)Y(y) = 0 \Rightarrow X'(0) = 0 \\ X'(a)Y(y) = 0 \Rightarrow X'(a) = 0 \end{cases}$$

$$\begin{cases} X'' + \lambda X = 0 \\ X'(0) = X'(a) = 0 \end{cases}$$

①. $\lambda = 0, X = Ax + B.$

$$X'(0) = A = 0 \Rightarrow X_0 = B_0.$$

②. $\lambda < 0, X = A \cdot e^{\sqrt{-\lambda}x} + B \cdot e^{-\sqrt{-\lambda}x}$

$$X'(x) = \sqrt{-\lambda} \cdot A \cdot e^{\sqrt{-\lambda}x} - \sqrt{-\lambda} \cdot B \cdot e^{-\sqrt{-\lambda}x}$$

$$X'(0) = \sqrt{-\lambda} \cdot A - \sqrt{-\lambda} \cdot B = 0 \Rightarrow X \equiv 0.$$

③. $\lambda > 0, X = A \cos \sqrt{\lambda}x + B \sin(\sqrt{\lambda}x).$

$$X' = -A \cdot \sqrt{\lambda} \cdot \sin(\sqrt{\lambda}x) + \sqrt{\lambda} B \cdot \cos(\sqrt{\lambda}x).$$

$$X'(0) = \sqrt{\lambda} B = 0, X'(a) = -A \cdot \sqrt{\lambda} \cdot \sin(\sqrt{\lambda}a) = 0.$$

$$\sqrt{\lambda}a = n\pi, \lambda_n = \left(\frac{n\pi}{a}\right)^2, n=1, \dots \Rightarrow \lambda_0 = 0.$$

$$\Rightarrow X_n = A_n \cos\left(\frac{n\pi}{a}x\right), n=1, \dots, X_0 = A_0.$$

$$\lambda \geq 0 \Rightarrow Y_n'' - \lambda Y_n = 0.$$

①. $\lambda = 0, Y'' = 0, \Rightarrow Y = C_0 y + D_0.$

②. $\lambda > 0, Y = C \cdot e^{\sqrt{\lambda}y} + D \cdot e^{-\sqrt{\lambda}y}$

$$\Rightarrow Y_n = B_n \cdot e^{\frac{n\pi}{a}y} + C_n \cdot e^{-\frac{n\pi}{a}y}, n=1, \dots$$

$$Y_0 = B_0 y + C_0.$$

$$X_n Y_n = A_n \cdot \cos\left(\frac{n\pi}{a}x\right) [B_n \cdot e^{\frac{n\pi}{a}y} + C_n \cdot e^{-\frac{n\pi}{a}y}]$$

$$= \cos\left(\frac{n\pi}{a}x\right) [A_n \cdot e^{\frac{n\pi}{a}y} + B_n \cdot e^{-\frac{n\pi}{a}y}]$$

$$X_0 Y_0 = A_0 (B_0 y + C_0) = A_0 y + B_0.$$

$$\Rightarrow U(x, y) = \sum_{n=0}^{\infty} X_n Y_n = A_0 y + B_0 + \sum_{n=1}^{\infty} \cos\left(\frac{n\pi}{a} x\right) [A_n \cdot e^{\frac{n\pi}{a} y} + B_n \cdot e^{-\frac{n\pi}{a} y}]$$

$$U(x, 0) = B_0 + \sum_{n=1}^{\infty} (A_n + B_n) \cos\left(\frac{n\pi}{a} x\right) = f(x).$$

边界条件:

$$f(x) = B_0 + \sum_{n=1}^{\infty} (A_n + B_n) \cos\left(\frac{n\pi}{a} x\right) = A_0 b + B_0 + \sum_{n=1}^{\infty} \cos\left(\frac{n\pi}{a} x\right) \cdot [A_n \cdot e^{\frac{n\pi}{a} b} + B_n \cdot e^{-\frac{n\pi}{a} b}] = g(x)$$

$$B_0 + \sum_{n=1}^{\infty} (A_n + B_n) \cos\left(\frac{n\pi}{a} x\right) = A_0 b + B_0 = \frac{1}{a} \int_0^a g(x) dx.$$

$$A_n \cdot e^{\frac{n\pi}{a} b} + B_n \cdot e^{-\frac{n\pi}{a} b} = \frac{2}{a} \int_0^a g(x) \cos\left(\frac{n\pi}{a} x\right) dx.$$

$$\int_0^a B_0 dx = \sum_{n=1}^{\infty} (A_n + B_n) \int_0^a \cos\left(\frac{n\pi}{a} x\right) dx$$

$$\Rightarrow A_0 = \frac{1}{ab} \int_0^a g(x) dx - \frac{1}{b} B_0$$

$$= \int_0^a f(x) dx.$$

$$= \frac{1}{ab} \int_0^a g(x) dx - \frac{1}{ab} \int_0^a f(x) dx.$$

$$n \geq 1, \int_0^a \cos\left(\frac{n\pi}{a} x\right) dx = 0 \Rightarrow U(x, y) = \frac{y}{ab} \int_0^a g(x) dx - \frac{y}{ab} \int_0^a f(x) dx$$

$$\Rightarrow \int_0^a f(x) dx = B_0 \cdot a.$$

$$+ \frac{1}{a} \int_0^a f(x) dx + \sum_{n=1}^{\infty} \cos\left(\frac{n\pi}{a} x\right) \left[\frac{2}{a} \int_0^a g(x) \cos\left(\frac{n\pi}{a} x\right) dx \right].$$

$$B_0 \int_0^a \cos\left(\frac{n\pi}{a} x\right) dx + \sum_{n=1}^{\infty} (A_n + B_n) \int_0^a \cos\left(\frac{n\pi}{a} x\right) \cos\left(\frac{m\pi}{a} x\right) dx.$$

$$= \sum_{n=1}^{\infty} (A_n + B_n) \int_0^a \cos\left(\frac{n\pi}{a} x\right) \cos\left(\frac{m\pi}{a} x\right) dx.$$

$$= \int_0^a f(x) \cos\left(\frac{m\pi}{a} x\right) dx.$$

(n > 1)

$$\text{余弦函数正交性: } \int_0^a \cos\left(\frac{n\pi}{a} x\right) \cos\left(\frac{m\pi}{a} x\right) dx = \begin{cases} 0 & n \neq m \\ \frac{a}{2} & n = m \geq 1 \end{cases}$$

$$\Rightarrow \frac{a}{2} (A_n + B_n) = \int_0^a f(x) \cos\left(\frac{m\pi}{a} x\right) dx.$$

$$= \int_0^a f(x) \cos\left(\frac{n\pi}{a} x\right) dx. \quad (\text{因为 } m = n).$$

(p5) ②. N-H-BC. Lap.
(完全(D)).

$$\begin{cases} U_{xx} + U_{yy} = 0 \\ U(0, y) = U(a, y) = 0 \\ U(x, 0) = p(x), U(x, b) = q(x) \end{cases}$$

$$[\text{解}] U = X Y. \quad X'' Y = -X Y'' \quad \frac{X''}{X} = -\frac{Y''}{Y} = -\lambda.$$

$$\begin{cases} X'' + \lambda X = 0 \\ Y'' - \lambda Y = 0 \end{cases} \quad \begin{cases} X(0) Y(y) = 0 \Rightarrow X(0) = 0 \\ X(a) Y(y) = 0 \Rightarrow X(a) = 0 \end{cases}$$

$$\begin{cases} X'' + \lambda X = 0 \\ X(0) = X(a) = 0 \end{cases} \quad \text{①. } \lambda = 0. \quad X = Ax + B.$$

$$X(0) = B \Rightarrow X \equiv 0.$$

②. $\lambda < 0$. $X \equiv 0$.

③. $\lambda > 0$. $X = A \cos(\sqrt{\lambda} x) + B \sin(\sqrt{\lambda} x)$.

$X(0) = A = 0$. $X(a) = B \sin(\sqrt{\lambda} a) = 0$. $\sqrt{\lambda} a = n\pi$. $\lambda_n = (\frac{n\pi}{a})^2$, $n = 1, \dots$

$\Rightarrow X_n = A_n \sin(\frac{n\pi}{a} x)$.

$Y'' - \lambda Y = 0$.

$\lambda > 0$. $Y_n = B_n \cdot e^{\frac{n\pi}{a} y} + C_n \cdot e^{-\frac{n\pi}{a} y}$, $n = 1, \dots$

$\Rightarrow X_n Y_n = \sin(\frac{n\pi}{a} x) [A_n \cdot e^{\frac{n\pi}{a} y} + B_n \cdot e^{-\frac{n\pi}{a} y}]$, $n = 1, \dots$

$\Rightarrow u(x, y) = \sum_{n=1}^{\infty} X_n Y_n = \sum_{n=1}^{\infty} \sin(\frac{n\pi}{a} x) [A_n \cdot e^{\frac{n\pi}{a} y} + B_n \cdot e^{-\frac{n\pi}{a} y}]$, $n = 1, \dots$

$u(x, y) = \sum_{n=1}^{\infty} \sin(\frac{n\pi}{a} x) [A_n \cdot e^{\frac{n\pi}{a} y} + B_n \cdot e^{-\frac{n\pi}{a} y}]$

$u(x, 0) = \sum_{n=1}^{\infty} \sin(\frac{n\pi}{a} x) \cdot [A_n + B_n] = \varphi(x)$.

$\Rightarrow A_n + B_n = \frac{2}{a} \int_0^a \varphi(x) \sin(\frac{n\pi}{a} x) dx$.

$u(x, b) = \sum_{n=1}^{\infty} \sin(\frac{n\pi}{a} x) [A_n \cdot e^{\frac{n\pi}{a} b} + B_n \cdot e^{-\frac{n\pi}{a} b}] = \psi(x)$

$\Rightarrow A_n \cdot e^{\frac{n\pi}{a} b} + B_n \cdot e^{-\frac{n\pi}{a} b} = \frac{2}{a} \int_0^a \psi(x) \sin(\frac{n\pi}{a} x) dx$.

$A_n \cdot e^{\frac{n\pi}{a} b} + B_n \cdot e^{-\frac{n\pi}{a} b} = \frac{2}{a} \int_0^a \varphi(x) \cdot e^{\frac{n\pi}{a} b} \cdot \sin(\frac{n\pi}{a} x) dx$

$B_n (e^{\frac{n\pi}{a} b} - e^{-\frac{n\pi}{a} b}) = \frac{2}{a} \int_0^a (\varphi(x) - \psi(x)) \cdot e^{\frac{n\pi}{a} b} \cdot \sin(\frac{n\pi}{a} x) dx$

$\Rightarrow B_n = \frac{\frac{2}{a} \int_0^a (\varphi(x) - \psi(x)) \cdot e^{-\frac{n\pi}{a} b} \sin(\frac{n\pi}{a} x) dx}{1 - e^{-\frac{2n\pi b}{a}}}$

$\Rightarrow A_n = \frac{\frac{2}{a} \int_0^a (\varphi(x) - \psi(x)) \cdot e^{-\frac{n\pi}{a} b} \sin(\frac{n\pi}{a} x) dx}{1 - e^{-\frac{2n\pi b}{a}}}$

③. N-H-B.C. Lap.
(π 界).

$\begin{cases} u_{xx} + u_{yy} = 0, & x \in (0, a), y \in (0, \infty) \\ u(0, y) = u(a, y) = 0 \\ u(x, 0) = \varphi(x), & u(x, \infty) = 0. \end{cases}$

[解]. $u = X Y$. $X'' Y = -X Y''$. $\frac{X''}{X} = -\frac{Y''}{Y} = -\lambda$.

$\begin{cases} X'' + \lambda X = 0 \\ Y'' - \lambda Y = 0 \end{cases} \quad \begin{cases} X(0) Y(y) = 0 \Rightarrow X(0) = 0 \\ X(a) Y(y) = 0 \Rightarrow X(a) = 0. \end{cases}$

$\begin{cases} X'' + \lambda X = 0 \\ X(0) = X(a) = 0. \end{cases} \quad \textcircled{1} \lambda = 0. X = Ax + B.$
 $X(0) = B = 0, X(a) = Aa + B = 0 \Rightarrow A = 0 \Rightarrow X \equiv 0$.

② $\lambda < 0$. ③ $\lambda > 0$. $X = A \cos \sqrt{\lambda} x + B \sin \sqrt{\lambda} x$.

$X(0) = A = 0$. $X(a) = B \sin \sqrt{\lambda} a = 0$. $\sqrt{\lambda} a = n\pi$,

$\lambda_n = (\frac{n\pi}{a})^2$, $n=1, \dots$. $X_n = A_n \sin(\frac{n\pi}{a} x)$, $n=1, \dots$

$\lambda > 0 \Rightarrow Y'' - \lambda Y = 0$.

$Y_n = B_n \cdot e^{\sqrt{\lambda} y} + C_n \cdot e^{-\sqrt{\lambda} y}$

$= B_n \cdot e^{\frac{n\pi}{a} y} + C_n \cdot e^{-\frac{n\pi}{a} y}$, $n=1, \dots$

$X_n Y_n = (A_n \cdot e^{\frac{n\pi}{a} y} + B_n \cdot e^{-\frac{n\pi}{a} y}) \cdot \sin(\frac{n\pi}{a} x)$, $n=1, \dots$

$U = \sum_{n=1}^{\infty} U_n = \sum_{n=1}^{\infty} \sin(\frac{n\pi}{a} x) (A_n \cdot e^{\frac{n\pi}{a} y} + B_n \cdot e^{-\frac{n\pi}{a} y})$

$U(x, \infty) = 0 \Rightarrow A_n = 0$.

$U(x, 0) = \sum_{n=1}^{\infty} \sin(\frac{n\pi}{a} x) \cdot B_n = f(x)$

$B_n = \frac{2}{a} \int_0^a f(x) \sin(\frac{n\pi}{a} x) dx$. $\square$

2. 极坐标 Lap.

$\begin{cases} \rho = \sqrt{x^2 + y^2} \\ \theta = \arctan \frac{y}{x} \end{cases} \Rightarrow \Delta U = U_{\rho\rho} + \frac{1}{\rho} U_{\rho} + \frac{1}{\rho^2} U_{\theta\theta}$

$\begin{cases} U_{yy} + U_{xx} = 0, x^2 + y^2 < \rho_0^2 \\ U(x, y) = f(x, y), x^2 + y^2 = \rho_0^2 \end{cases} \Rightarrow \begin{cases} U_{\rho\rho} + \frac{1}{\rho} U_{\rho} + \frac{1}{\rho^2} U_{\theta\theta} = 0 \\ U(\rho_0, \theta) = f(\theta) \end{cases}$

① 自然 BC. $|u(0, \theta)| < +\infty$

② 周期 BC. $u(\rho, \theta) = u(\rho, \theta + 2\pi)$.

p13) ①. 极坐标解 $\begin{cases} U_{yy} + U_{xx} = 0 \\ U(x, y) = f(x, y) \end{cases}$. 圆.

[解] $\Rightarrow \begin{cases} U_{\rho\rho} + \frac{1}{\rho} U_{\rho} + \frac{1}{\rho^2} U_{\theta\theta} = 0 \\ U(\rho_0, \theta) = f(\theta) \end{cases}$

$U(\rho, \theta) = P(\rho) \theta(\theta)$.

$P''\theta + \frac{1}{\rho} P'\theta + \frac{1}{\rho^2} P\theta'' = 0$.

$(P'' + \frac{1}{\rho} P')\theta = -\frac{1}{\rho^2} P\theta''$

$-\frac{\theta''}{\theta} = \frac{r^2}{P} \frac{P'' + rP'}{P} = (\lambda)$

$\begin{cases} \theta'' + \lambda\theta = 0, \theta \in [0, 2\pi] \\ \text{周期} \end{cases}$

$\theta(0) = \theta(0 + 2\pi)$

① $\lambda = 0$. $\theta = A\theta + B$. $\theta(0) = B = \theta(2\pi) = A \cdot 2\pi + B \Rightarrow A = 0$
 $\Rightarrow \theta_0 = B_0$

$$②. \lambda < 0. \theta = A \cdot e^{\sqrt{\lambda}\theta} + B \cdot e^{-\sqrt{\lambda}\theta} \Rightarrow \theta = 0.$$

$$③. \lambda > 0. \theta = A \cos \sqrt{\lambda}\theta + B \sin \sqrt{\lambda}\theta.$$

$$\theta(0) = A = \theta(2\pi) = A \cos(\sqrt{\lambda} \cdot 2\pi) + B \sin(\sqrt{\lambda} \cdot 2\pi)$$

$$\sqrt{\lambda} \cdot 2\pi = 2n\pi, \lambda = n^2, n=1, \dots, \lambda=0.$$

$$\Rightarrow \theta_n = A_n \cos(n\theta) + B_n \sin(n\theta), n=1, \dots$$

$$\theta_0 = A_0.$$

$$\rho^2 \rho'' + \rho \rho' - \lambda \rho = 0. \rightarrow \text{Euler 方程}.$$

$$(\text{Euler 法}), \text{ 设 } \rho = e^t, t = \ln \rho.$$

$$\rho' = \frac{d\rho}{dt} = \frac{d\rho}{dt} \cdot \frac{dt}{d\rho} = \frac{1}{\rho} \cdot \frac{d\rho}{dt}$$

$$\rho'' = \frac{d}{d\rho}(\rho') = \frac{d}{d\rho} \left(\frac{1}{\rho} \cdot \frac{d\rho}{dt} \right)$$

$$= -\frac{1}{\rho^2} \cdot \frac{d\rho}{dt} + \frac{1}{\rho} \cdot \frac{d}{d\rho} \cdot \frac{d\rho}{dt}$$

$$\left(\frac{d}{d\rho} \cdot \frac{d\rho}{dt} = \frac{d^2\rho}{dt^2} \cdot \frac{dt}{d\rho} = \frac{d^2\rho}{dt^2} \cdot \frac{1}{\rho} \right)$$

$$= -\frac{1}{\rho^2} \cdot \frac{d\rho}{dt} + \frac{1}{\rho^2} \cdot \frac{d^2\rho}{dt^2}$$

$$= \frac{1}{\rho^2} \left(\frac{d^2\rho}{dt^2} - \frac{d\rho}{dt} \right).$$

$$\Rightarrow \frac{d^2\rho}{dt^2} - \frac{d\rho}{dt} + \frac{d\rho}{dt} - \lambda \rho = 0.$$

$$\Rightarrow \begin{cases} \rho_{tt} - \lambda \rho = 0 \\ \text{自然 BC. } |u(0, +\infty)| < +\infty \end{cases}$$

$$①. \lambda = 0. \rho_0 = A_0 t + B_0 = A_0 \ln \rho + B_0.$$

$$\rho \rightarrow 0, |A_0 \ln \rho + B_0| \rightarrow \infty \Rightarrow A_0 = 0 \Rightarrow \rho_0 = B_0.$$

$$③. \lambda > 0. \rho = A \cdot e^{\sqrt{\lambda}t} + B \cdot e^{-\sqrt{\lambda}t}$$

$$\rho_n = A_n \cdot e^{n \cdot \ln \rho} + B_n \cdot e^{-n \cdot \ln \rho}$$

$$= A_n \cdot \rho^n + B_n \cdot \rho^{-n}$$

$$\rho \rightarrow 0, |\rho^{-n}| \rightarrow +\infty \Rightarrow B_n = 0 \Rightarrow \rho_n = A_n \cdot \rho^n, n=1, \dots$$

$$\Rightarrow u = \sum_{n=0}^{\infty} \theta_n \rho_n = C_0 \cdot A_0 + \sum_{n=1}^{\infty} (C_n \cdot \rho^n (A_n \cos(n\theta) + B_n \sin(n\theta)))$$

$$= A_0 + \sum_{n=1}^{\infty} \rho^n (A_n \cos(n\theta) + B_n \sin(n\theta))$$

$$\Rightarrow A_0 = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) d\theta, A_n = \frac{1}{\pi \rho_0^n} \int_0^{2\pi} f(\theta) \cos(n\theta) d\theta, B_n = \frac{1}{\pi \rho_0^n} \int_0^{2\pi} f(\theta) \sin(n\theta) d\theta$$

$$u(\rho, \theta) = A_0 + \sum_{n=1}^{\infty} \rho_0^n (A_n \cos(n\theta) + B_n \sin(n\theta)) = f(\theta)$$

* $n=0$ 时各系数为 $\frac{1}{2}$,

40 $\pi, \frac{1}{t}, \dots$

(见 P42 例 3).

$$\int_0^{2\pi} \cos(n\theta) \sin(m\theta) d\theta = 0.$$

$$\int_0^{2\pi} \sin(n\theta) \sin(m\theta) d\theta = \begin{cases} 0, & n \neq m. \\ \pi, & n = m. \end{cases}$$

$$\Rightarrow \rho_0^n \cdot B_n = \int_0^{2\pi} f(\theta) \sin(n\theta) d\theta$$

5.

②. 极坐标解. 17.

$$\begin{cases} \frac{1}{\rho} \cdot \frac{\partial}{\partial \rho} \left(\rho \cdot \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \cdot \frac{\partial^2 u}{\partial \theta^2} = 0 & \rho \in (a, b), \theta \in [0, 2\pi] \\ u(a, \theta) = 0, u(b, \theta) = 1 \end{cases}$$

[解]. $\frac{1}{\rho} \cdot \frac{\partial}{\partial \rho} (\rho \cdot u_\rho) + \frac{1}{\rho^2} \cdot u_{\theta\theta} = 0$

$$\frac{1}{\rho} \cdot u_\rho + u_{\rho\rho} + \frac{1}{\rho^2} u_{\theta\theta} = 0$$

$$u = P(\rho) \cdot \theta(\theta), u_\rho = P' \theta, u_{\rho\rho} = P'' \theta, u_{\theta\theta} = P \theta'',$$

$$\frac{1}{\rho} \cdot P' \theta + P'' \theta = - \frac{1}{\rho^2} P \theta''$$

$$\frac{\rho P' + \rho^2 P''}{\rho} = - \frac{\theta''}{\theta} = \lambda$$

$$\rho^2 P'' + \rho P' - \lambda P = 0, \theta'' + \lambda \theta = 0$$

$$\begin{cases} \theta'' + \lambda \theta = 0 \\ \text{周期 } B.C. \end{cases}$$

$$\theta(\theta) = \theta(\theta + 2\pi)$$

①. $\lambda = 0, \theta = A\theta + B, \theta(0) = B = \theta(2\pi) = A \cdot 2\pi + B \Rightarrow A = 0 \Rightarrow \theta_0 = B_0$

②. $\lambda < 0, \theta = A \cdot e^{\sqrt{-\lambda}\theta} + B \cdot e^{-\sqrt{-\lambda}\theta}, \theta \equiv 0$

③. $\lambda > 0, \theta = A \cos(\sqrt{\lambda}\theta) + B \sin(\sqrt{\lambda}\theta)$

$$\theta(0) = A = \theta(2\pi) = A \cos(2\pi \cdot \sqrt{\lambda}) = B \sin(2\pi \sqrt{\lambda})$$

$$\begin{cases} \sin(2\pi \cdot \sqrt{\lambda}) = 0 \\ \cos(2\pi \cdot \sqrt{\lambda}) = 1 \end{cases} \quad \begin{matrix} 2\pi \cdot \sqrt{\lambda} = n \cdot 2\pi \\ \lambda = n^2, n = 1, \dots, \lambda_0 > 0 \end{matrix}$$

$$\Rightarrow \theta_n = A_n \cdot \cos(\pi n) + B_n \cdot \sin(\pi n), n = 1, \dots, \theta_0 = B_0$$

$$\rho^2 P'' + \rho P' - \lambda P = 0$$

由 Euler, $t = \ln \rho \Rightarrow \begin{cases} P_{tt} - \lambda P = 0 \\ |P(\rho)| < +\infty \end{cases}$

①. $\lambda = 0, P = C_0 t + D_0, \lambda_0 = 0, P_0 = C_0 t + D_0 = C_0 \cdot \ln \rho + D_0$

②. $\lambda > 0, P = C \cdot e^{\sqrt{\lambda}t} + D \cdot e^{-\sqrt{\lambda}t}$
 $= C \cdot e^{n \ln \rho} + D \cdot e^{-n \ln \rho}$

$$\Rightarrow P_n = C_n \cdot \rho^n + D_n \cdot \rho^{-n}, n = 1, \dots$$

$$\Rightarrow u = \sum_{n=0}^{\infty} P_n \cdot \theta = B_0 \cdot (C_0 \cdot \ln \rho + D_0) + \sum_{n=1}^{\infty} (C_n \cdot \rho^n + D_n \cdot \rho^{-n}) (A_n \cdot \cos(\pi n))$$

$$= B_0 \cdot C_0 \cdot \ln \rho + B_0 D_0 + \sum (A_n (C_n \cdot \rho^n + D_n \cdot \rho^{-n}) + B_n \cdot \sin(\pi n))$$

$$= C_0 \ln \rho + D_0 + \sum [(A_n \cdot \rho^n + B_n \cdot \rho^{-n}) \cos(\pi n) + (C_n \cdot \rho^n + D_n \cdot \rho^{-n}) \cdot \sin(\pi n)]$$

$$u(a, \theta) = C_0 \ln a + D_0 + \sum (A_n \cdot a^n + B_n \cdot a^{-n}) \cos(\pi n) + (C_n \cdot a^n + D_n \cdot a^{-n}) \cdot \sin(\pi n) = 0$$

$$u(b, \theta) = C_0 \ln b + D_0 + \sum (A_n \cdot b^n + B_n \cdot b^{-n}) \cos(\pi n) + (C_n \cdot b^n + D_n \cdot b^{-n}) \cdot \sin(\pi n) = 1$$

$$\Rightarrow \begin{cases} C_0 \ln a + D_0 = 0 \\ C_0 \ln b + D_0 = 1 \end{cases} \quad \begin{cases} A_n \cdot a^n + B_n \cdot a^{-n} = 0 \\ A_n \cdot b^n + B_n \cdot b^{-n} = 0 \end{cases} \quad \begin{cases} C_n \cdot a^n + D_n \cdot a^{-n} = 0 \\ C_n \cdot b^n + D_n \cdot b^{-n} = 0 \end{cases}$$

$$C_0 \ln b - C_0 \ln a = C_0 \ln \frac{b}{a} = 1, \quad C_0 = \left(\ln \frac{b}{a} \right)^{-1}, \quad D_0 = -C_0 \ln a = -\frac{\ln a}{\ln \frac{b}{a}}$$

$$A_n \cdot a^n b^n + B_n \cdot a^{-n} b^{-n} = 0 \Rightarrow A_n = B_n = 0, \quad C_n = D_n = 0$$

$$A_n \cdot a^n b^n + B_n \cdot a^n b^{-n} = 0 \Rightarrow A_n = B_n = 0, \quad C_n = D_n = 0$$

$$\Rightarrow u = \left(\ln \frac{b}{a} \right)^{-1} \cdot \ln p = \frac{\ln a}{\ln \frac{b}{a}}$$

$$= \frac{\ln p - \ln a}{\ln \frac{b}{a}} = \frac{\ln \frac{p}{a}}{\ln \frac{b}{a}}$$

③. 极坐标解. 原.

$$\begin{cases} \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0, \quad r < 1, \quad \theta \in [0, \frac{\pi}{3}] \\ u(1, \theta) = \sin \theta, \quad u(r, 0) = u(r, \frac{\pi}{3}) = 0 \end{cases}$$

[解]. $r^2 \theta'' + r \theta' + \theta^2 = 0$

$$\frac{r^2 \theta'' + r \theta'}{r} = -\frac{\theta^2}{\theta} = \lambda$$

$r^2 \theta'' + r \theta' - \lambda \theta = 0, \quad \textcircled{1} \lambda = 0, \quad \theta = A\theta + B.$

$$\begin{cases} \theta'' + \lambda \theta = 0 \\ \theta(0) = \theta(\frac{\pi}{3}) = 0 \end{cases} \quad \begin{aligned} \theta(0) = B = \frac{\pi}{3} A + B \Rightarrow A = 0 \\ \theta_0 = B_0. \end{aligned}$$

$\textcircled{2} \lambda < 0, \quad \theta \equiv 0.$

$\textcircled{3} \lambda > 0, \quad \theta = A \cdot \cos(\sqrt{\lambda} \theta) + B \cdot \sin(\sqrt{\lambda} \theta)$

$$\theta(0) = A = A \cdot \cos(\sqrt{\lambda} \cdot \frac{\pi}{3}) + B \cdot \sin(\sqrt{\lambda} \cdot \frac{\pi}{3}) = 0, \quad A = 0.$$

$$\frac{\pi}{3} \cdot \sqrt{\lambda} = n\pi, \quad \lambda = 9n^2, \quad n = 1, \dots$$

$$\theta_n = A_n \cdot \sin(3n\theta), \quad n = 1, \dots, \quad \theta_0 = A_0.$$

Euler. $t = \ln p \Rightarrow \begin{cases} P_{tt} - \lambda P = 0 \\ |P(p)| < +\infty \end{cases}$

$\textcircled{1} \lambda = 0, \quad P = Bt + C = B \cdot \ln p + C$

$$\ln p \rightarrow \infty (p \rightarrow 0) \Rightarrow B = 0 \Rightarrow P_0 = C_0.$$

$\textcircled{2} \lambda > 0, \quad P = B \cdot e^{\sqrt{\lambda} t} + C \cdot e^{-\sqrt{\lambda} t}$

$$P_n = B_n \cdot e^{3n \ln p} + C_n \cdot e^{-3n \ln p}$$

$$= B_n \cdot p^{3n} + C_n \cdot p^{-3n}$$

$$= B_n \cdot p^{3n}, \quad n = 1, \dots, \quad P_0 = C_0.$$

$$\Rightarrow u = A_0 \cdot C_0 + \sum_{n=1}^{\infty} A_n \cdot \sin(3n\theta) \cdot B_n \cdot p^{3n}$$

$$= B_0 + \sum_{n=1}^{\infty} 4n \cdot p^{3n} \cdot \sin(3n\theta)$$

$$u(1, \theta) = B_0 + \sum A_n \cdot \sin(3n\theta) = \sin 6\theta$$

$$\Rightarrow B_0 = 0, A_n = \begin{cases} 1, & n=2 \\ 0, & n \neq 2 \end{cases}$$

$$\Rightarrow u = \rho^6 \cdot \sin(6\theta).$$

□

3. Lap / 实域 / 拉氏变换. (在 $[0, +\infty)$ 有定义).

$$\mathcal{L}[f(x)] = \int_0^{+\infty} f(x) \cdot e^{-sx} dx = \tilde{f}(s)$$

$$\mathcal{L}^{-1}[\tilde{f}(s)] = \frac{1}{2\pi i} \int_{\beta-i\infty}^{\beta+i\infty} \tilde{f}(s) \cdot e^{sx} ds = f(x).$$

①. 线性. $\mathcal{L}[\alpha f + \beta g] = \alpha \mathcal{L}[f] + \beta \mathcal{L}[g].$

②. $\mathcal{L}[f'] = s \mathcal{L}[f] - f(0).$

$$\begin{aligned} \mathcal{L}[f^{(n)}] &= s \cdot \mathcal{L}[f^{(n-1)}] - f^{(n-1)}(0) \\ &= s^2 \mathcal{L}[f^{(n-2)}] - s f^{(n-2)}(0) - f^{(n-1)}(0) \\ &= \dots = s^n \mathcal{L}[f] - s^{n-1} f(0) - \dots - s f^{(n-2)}(0) - f^{(n-1)}(0). \end{aligned}$$

③. $\mathcal{L}[e^{ax} \cdot f(x)] = \tilde{f}(s-a)$
 $\mathcal{L}^{-1}[\tilde{f}(s-a)] = e^{ax} \cdot f(x).$

④. 卷积定理.

$$\begin{aligned} \mathcal{L}[f * g(x)] &= \mathcal{L}[f(x)] \cdot \mathcal{L}[g(x)] \\ \mathcal{L}^{-1}[\mathcal{L}[f(x)] \mathcal{L}[g(x)]] &= (f * g)(x). \end{aligned}$$

II. Lap 解 $(H-BC)$. N-H-IC. H-E. 波.

$$\begin{cases} u_{tt} = u_{xx}, & x \in (0, 1), t > 0. \\ u(0, t) = u(1, t) = 0. \\ u(x, 0) = \sin \pi x, & u_t(x, 0) = -\sin \pi x. \end{cases}$$

[解]. 对 t 作 lap 变换.

$$\mathcal{L}[u(x, t)] = U(x, s).$$

$$\begin{aligned} \Rightarrow \mathcal{L}[u_{tt}] &= s^2 \mathcal{L}[u] - s \cdot u(x, 0) - u_t(x, 0) \\ &= s^2 U - s \cdot \sin(\pi x) + \sin(\pi x) \\ &= s^2 U + (1-s) \sin(\pi x). \end{aligned}$$

$$\mathcal{L}[u_{xx}] = \{\mathcal{L}[u]\}_{xx} = U_{xx}.$$

$$\Rightarrow s^2 U + (1-s) \sin(\pi x) = U_{xx}$$

$$BC \text{ 作lap. } \begin{cases} \mathcal{L}[u(0,t)] = U(0,s) = \mathcal{L}[0] = 0, \\ \mathcal{L}[u(1,t)] = U(1,s) = 0. \end{cases}$$

$$\Rightarrow \begin{cases} U_{xx} - s^2 U + (s-1) \sin(\pi x) = 0, \\ U(0,s) = 0 = U(1,s). \end{cases}$$

$$U_{xx} - s^2 U = 0 \Rightarrow U_h = A \cdot e^{sx} + B \cdot e^{-sx}$$

$$\text{设 } U_p = C \cdot \sin(\pi x) \Rightarrow C = \frac{s-1}{s^2 + \pi^2}$$

$$\Rightarrow U = A \cdot e^{sx} + B \cdot e^{-sx} + \frac{s-1}{s^2 + \pi^2} \cdot \sin(\pi x)$$

$$U(0,s) = A + B = 0 = A \cdot e^s + B \cdot e^{-s}$$

$$\Rightarrow A = B = 0.$$

$$\Rightarrow U(x,s) = \frac{s-1}{s^2 + \pi^2} \cdot \sin(\pi x)$$

$$\Rightarrow u(x,t) = \mathcal{L}^{-1}[U(x,s)] = \mathcal{L}^{-1}\left[\frac{s-1}{s^2 + \pi^2} \cdot \sin(\pi x)\right].$$

b. Lap 解 N-H-Bc. N-H-IC. H-E 型.

$$\begin{cases} u_t = u_{xx}, & x \in (0, L), t > 0 \\ u(x, 0) = u_0 \\ u_x(0, t) = 0, u(L, t) = -u_1. \end{cases}$$

[解], 对 t 作 lap. $\mathcal{L}[u(x,t)] = U(x,s).$

$$\mathcal{L}[u_t] = s \cdot \mathcal{L}[u] - u(x, 0) = s \cdot U - u_0 = U_{xx}$$

$$\mathcal{L}[u_x(0,t)] = 0 = U_x(0,s)$$

$$\mathcal{L}[u(L,t)] = U(L,s) = \mathcal{L}[-u_1] = \frac{-u_1}{s} \rightarrow \text{常数作lap} = \frac{1}{s} \cdot \text{常数}.$$

$$\begin{cases} U_{xx} - s \cdot U + u_0 = 0 \\ U_x(0,s) = 0, U(L,s) = -\frac{u_1}{s} \end{cases}$$

$$U_h = A \cdot e^{\sqrt{s}x} + B \cdot e^{-\sqrt{s}x}.$$

$$\text{设 } U_p = C \cdot u_0 \Rightarrow C = \frac{1}{s}.$$

$$\Rightarrow U = A \cdot e^{\sqrt{s}x} + B \cdot e^{-\sqrt{s}x} + \frac{u_0}{s}.$$

$$\Rightarrow u(x,t) = \mathcal{L}^{-1}[U(x,s)].$$

4. 特殊.

①. 混合偏导 $\rightarrow$ 2次积分转换为2个单变量函数相加.

$$\begin{cases} u_{xy}(x,y) = x^2y & x>1, y>0 \\ u(x,0) = x^3, u(1,y) = \cos y. \end{cases}$$

[解]. $\iint u_{xy}(x,y) dx dy = \iint x^2y dx dy$

$$\Rightarrow u = \frac{x^3y^2}{6} + f_1(x) + f_2(y)$$

$$\begin{cases} u(x,0) = x^3 \Rightarrow x^3 = f_1(x) + f_2(0) \\ u(1,y) = \cos y \Rightarrow \cos y = \frac{y^2}{6} + f_1(1) + f_2(y) \end{cases}$$

需求出 $f_2(0), f_1(1)$.

$$u(1,0) = \cos 1 = 1 = f_1(1) + f_2(0)$$

$$\Rightarrow u = \frac{x^3y^2}{6} + x^3 + \cos y - \frac{y^2}{6} - 1.$$

④. 此题也可 Lap 解).

②. 极限 BC $\rightarrow$ Frontier. (不用 Lap 因为无 $u_y(x,0)$).

思考: 若对 y 作 Lap, 则需 $u(x,0), u_y(x,0)$.

$$\begin{cases} u_{xx} + u_{yy} = 0, x \in \mathbb{R}, y > 0. \\ u(x,0) = \phi(x) \\ \lim_{x^2+y^2 \rightarrow \infty} u(x,y) = 0. \end{cases}$$

[解]. $\mathcal{F}[u(x,y)] = U(\zeta, y)$.

$$\mathcal{F}[u_{xx} + u_{yy}] = (\zeta^2) \cdot U + U_{yy} = 0$$

$$\mathcal{F}[u(x,0)] = U(\zeta, 0) = \mathcal{F}[\phi(x)] = \Phi(\zeta)$$

$$\mathcal{F}\left[\lim_{x^2+y^2 \rightarrow \infty} u(x,y)\right] = \lim_{y \rightarrow \infty} U = 0$$

$$\Rightarrow \begin{cases} U_{yy} - \zeta^2 U = 0 \\ U(\zeta, 0) = \Phi(\zeta) \\ \lim_{y \rightarrow \infty} U = 0 \end{cases}$$

$$U = A \cdot e^{\frac{1}{2}|\zeta|y} + B \cdot e^{-\frac{1}{2}|\zeta|y}$$

$$y \rightarrow \infty, |\zeta|y \rightarrow \infty \Rightarrow A = 0$$

$$u(\zeta, 0) = B = \Phi(\zeta)$$

$$\Rightarrow U = \Phi(\zeta) \cdot e^{-\frac{1}{2}|\zeta|y}$$

$$\Rightarrow u(x,y) = \mathcal{F}^{-1}[U(\zeta,y)] = \mathcal{F}^{-1}[\Phi(\zeta) \cdot e^{-\frac{1}{2}|\zeta|y}]$$

$$= \phi(x) * \mathcal{F}^{-1}[e^{-\frac{1}{2}|\zeta|y}]$$

$$= \phi(x) * \frac{y}{\pi(x^2+y^2)}$$

$$= \frac{y}{\pi} \cdot \int_{-\infty}^{+\infty} \frac{\phi(\zeta)}{(x-\zeta)^2+y^2} d\zeta.$$

④.

③. 非齐次项与 x, t 无关的 N-H-E $\rightarrow$ 变量代换让原问题齐次化.

$$\begin{cases} u_t = a^2 u_{xx} + u \\ u(0, t) = u(1, t) = 0 \\ u(x, 0) = \varphi(x) \end{cases}$$

[解]. 变量代换 $u = e^t \cdot v$

$$\Rightarrow e^t \cdot v_t + e^t v = e^t \cdot a^2 v_{xx} + e^t v$$

$$u(0, t) = e^t \cdot v(0, t) = 0$$

$$u(1, t) = e^t \cdot v(1, t) = 0$$

$$u(x, 0) = e^0 \cdot v(x, 0) = v(x, 0)$$

$$\Rightarrow \begin{cases} v_t = a^2 v_{xx} \\ v(0, t) = v(1, t) = 0 \\ v(x, 0) = \varphi(x) \end{cases} \quad \begin{array}{l} \text{简化为 H-E 热.} \\ \text{(此题也可直接分离变量).} \end{array}$$

*注: N-E 非齐次部分与 x, t 相关时 ($f(x, t)$), 不能变量代换, 只能分离变量, 如下:

④. $\begin{cases} V_{tt} = a^2 V_{xx} + f(x, t) \\ V(0, t) = V(1, t) = 0 \\ V(x, 0) = 0 = V_t(x, 0) \end{cases} \rightarrow$ 特征函数展开法:
先解出 H-E 的特征函数再设

[解]. $V = X T, X T'' = a^2 X'' T$

$$\frac{X''}{X} = \frac{T''}{a^2 T} = -\lambda \quad \begin{cases} X'' + \lambda X = 0 \\ T'' + \lambda a^2 T = 0 \end{cases}$$

$$X(0) T(t) = 0 \Rightarrow X(0) = 0 = X(1)$$

$$\begin{cases} X'' + \lambda X = 0 \\ X(0) = 0 = X(1) \end{cases} \quad \begin{array}{l} \text{①. } \lambda = 0, X = Ax + B \\ X(0) = b = A \cdot 0 + B \Rightarrow X \equiv 0 \end{array}$$

$$\text{②. } \lambda > 0, X = A \cos(\sqrt{\lambda} x) + B \sin(\sqrt{\lambda} x)$$

$$X(0) = A = A \cos(\sqrt{\lambda} \cdot 0) + B \sin(\sqrt{\lambda} \cdot 0)$$

$$\Rightarrow X_n = B_n \cdot \sin\left(\frac{n\pi}{l} x\right), n = 1, \dots$$

$$\sqrt{\lambda} l = n\pi, \lambda = \left(\frac{n\pi}{l}\right)^2, n = 1, \dots$$

$$\text{设 } V = \sum_{n=1}^{\infty} V_n(t) \cdot \sin \frac{n\pi}{l} x$$

$$\Rightarrow \sum V_n'' \cdot \sin\left(\frac{n\pi}{l} x\right) = a^2 \cdot \sum V_n(t) \cdot \sin\left(\frac{n\pi}{l} x\right) \cdot \left(-\frac{n^2\pi^2}{l^2}\right) + f(x, t)$$

$$\Rightarrow f(x, t) = \sum \left[\sin\left(\frac{n\pi}{l} x\right) \cdot \left(V_n'' + V_n \cdot \left(\frac{n\pi}{l}\right)^2\right) \right]$$

$$\text{记 } f_n(t) = V_n'' + V_n \cdot \left(\frac{n\pi}{l}\right)^2$$

$$\Rightarrow \begin{cases} f(x, t) = \sum \sin\left(\frac{n\pi}{l} x\right) \cdot f_n \\ V_n'' + V_n \cdot \left(\frac{n\pi}{l}\right)^2 - f_n = 0 \end{cases}$$

$$\text{由 } B.C., V(x, 0) = \sum V_n(0) \cdot \sin\left(\frac{n\pi}{l}x\right) = 0$$

$$\Rightarrow V_n(0) = 0$$

$$V_t(x, 0) = \sum V_n'(0) \cdot \sin\left(\frac{n\pi}{l}x\right) = 0$$

$$\Rightarrow V_n'(0) = 0$$

$$\Rightarrow \begin{cases} V_n'' + V_n \cdot \left(\frac{qn\pi}{l}\right)^2 - f_n = 0 \\ V_n(0) = 0, V_n'(0) = 0 \end{cases}$$

■

④ - b. $\begin{cases} u_t = a^2 u_{xx} + \sin \omega t & x \in [0, l] \\ u_x(0, t) = u_x(l, t) = 0 \\ u(x, 0) = 0 \end{cases}$

分离变量法: $u = X \cdot T, X \cdot T' = a^2 X'' \cdot T, \frac{X''}{X} = \frac{T'}{a^2 T} = -\lambda$

$$\begin{cases} X'' + \lambda X = 0 \\ X'(0) = X'(l) = 0 \end{cases} \quad \text{①. } \lambda = 0, X = Ax + B, X'(0) = A = 0 \Rightarrow X_0 = B_0, \lambda_0 = 0$$

$$\text{②. } \lambda < 0, X \equiv 0$$

$$\text{③. } \lambda > 0, X = A \cos(\sqrt{\lambda}x) + B \sin(\sqrt{\lambda}x)$$

$$X' = -A \sin(\sqrt{\lambda}x) \cdot \sqrt{\lambda} + B \sqrt{\lambda} \cdot \cos(\sqrt{\lambda}x)$$

$$X'(0) = B \sqrt{\lambda} = 0 \Rightarrow B = 0$$

$$X'(l) = -A \sqrt{\lambda} \cdot \sin(\sqrt{\lambda}l) = 0$$

$$\sqrt{\lambda}l = n\pi \Rightarrow \lambda = \left(\frac{n\pi}{l}\right)^2, n = 1, 2, \dots$$

$$\Rightarrow X_n = A_n \cdot \cos\left(\frac{n\pi}{l}x\right), n = 0, 1, 2, \dots$$

$$\text{设 } u(x, t) = \sum_{n=0}^{\infty} U_n(t) \cdot \cos\left(\frac{n\pi}{l}x\right)$$

$$\Rightarrow \sum U_n' \cdot \cos\left(\frac{n\pi}{l}x\right) = a^2 \cdot \sum_{n=0}^{\infty} U_n(t) \cdot \cos\left(\frac{n\pi}{l}x\right) \cdot \left(-\frac{n^2\pi^2}{l^2}\right) + \sin \omega t$$

$$\Rightarrow \sin \omega t = \sum \left[\cos\left(\frac{n\pi}{l}x\right) \cdot \left(U_n' + U_n \cdot \left(\frac{qn\pi}{l}\right)^2\right) \right]$$

$$\int \sin \omega t \cdot \cos\left(\frac{m\pi}{l}x\right) dx = \sum \int \left[U_n' + \left(\frac{qn\pi}{l}\right)^2 U_n \right] \cdot \cos\left(\frac{n\pi}{l}x\right) \cos\left(\frac{m\pi}{l}x\right) dx$$

$$\text{(比较系数)} \Rightarrow U_n' + \left(\frac{qn\pi}{l}\right)^2 U_n = \frac{2}{l} \int \sin \omega t \cdot \cos\left(\frac{n\pi}{l}x\right) dx$$

$$n = 0, U_0' = \frac{2}{l} \int_0^l \sin \omega t dx = \sin \omega t$$

$$n \geq 1, U_n' + \left(\frac{qn\pi}{l}\right)^2 U_n = \frac{2}{l} \int \sin \omega t \cdot \cos\left(\frac{n\pi}{l}x\right) dx = 0$$

$$u(x, 0) = 0 \Rightarrow U_n(0) = 0$$

$$\Rightarrow n = 0, U_0'(t) = \sin \omega t, U_0(0) = 0 \Rightarrow U_0(t) = \frac{1 - \cos \omega t}{\omega}$$

$$n \geq 1, U_n(0) = 0, U_n' + \left(\frac{qn\pi}{l}\right)^2 U_n = 0$$

$$\Rightarrow U_n(t) = C_n \cdot e^{-\left(\frac{qn\pi}{l}\right)^2 t}, C_n = 0 \Rightarrow U_n = 0$$

$$\Rightarrow u = \frac{1}{\omega} (1 - \cos \omega t) \cdot \cos\left(\frac{n\pi}{l}\right)$$

■

$$④-C. \begin{cases} u_{tt} = a^2 u_{xx} + \sin\left(\frac{2\pi x}{l}\right) \sin\left(\frac{2a\pi}{l}t\right), & x \in (0, l), t > 0 \\ u(0, t) = u(l, t) = 0 \\ u(x, 0) = u_t(x, 0) = 0. \end{cases}$$

H-B, H-IC, N-H-E 法

→ 特征函数展开法.
必须 BC 齐次才能用.

①. $u = XT, \quad XT'' = a^2 X''T, \quad \frac{X''}{X} = \frac{T''}{T} = -\lambda$

$$\begin{cases} X'' + \lambda X = 0 & ① \lambda = 0, X = Ax + B, \\ X(0) = X(l) = 0 & X(0) = B = A \cdot 0 + B \Rightarrow A = 0, X = B, \\ & X(l) = B = A \cdot l + B \Rightarrow A = 0, X = B. \end{cases}$$

③. $\lambda > 0, X = A \cos(\sqrt{\lambda}x) + B \sin(\sqrt{\lambda}x)$

$$X(0) = A = A \cos(\sqrt{\lambda} \cdot 0) + B \sin(\sqrt{\lambda} \cdot 0) = 0$$

$$\sqrt{\lambda}l = n\pi, \quad \lambda = \left(\frac{n\pi}{l}\right)^2, \quad n = 1, 2, \dots$$

$$\Rightarrow X_n = A_n \cdot \sin\left(\frac{n\pi}{l}x\right), \quad n = 0, 1, 2, \dots$$

$$\text{设 } u(x, t) = \sum_{n=0}^{\infty} U_n(t) \cdot \sin\left(\frac{n\pi}{l}x\right).$$

$$\sum U_n'' \cdot \sin\left(\frac{n\pi}{l}x\right) = \sum U_n \cdot \sin\left(\frac{m\pi}{l}x\right) \cdot \left(\frac{a^2 n^2 \pi^2}{l^2}\right) + \sin\left(\frac{2\pi x}{l}\right) \cdot \sin\left(\frac{2a\pi}{l}t\right)$$

$$\sin\left(\frac{2\pi x}{l}\right) \sin\left(\frac{2a\pi}{l}t\right) = \sum \left[\sin\left(\frac{n\pi}{l}x\right) \cdot (U_n'' + U_n \cdot \left(\frac{a^2 n^2 \pi^2}{l^2}\right)) \right]$$

$$\int_0^l \sin\left(\frac{2\pi x}{l}\right) \sin\left(\frac{2a\pi}{l}t\right) \sin\left(\frac{m\pi}{l}x\right) dx$$

(正交性) $\Rightarrow n=2, \begin{cases} U_2'' + U_2 \cdot \left(\frac{2a\pi}{l}\right)^2 = \sin\left(\frac{2a\pi}{l}t\right) \\ U_2(0) = 0, U_2'(0) = 0. \end{cases}$

$$n \neq 2, \begin{cases} U_n'' + U_n \cdot \left(\frac{a^2 n^2 \pi^2}{l^2}\right) = 0 \\ U_n(0) = 0, U_n'(0) = 0. \end{cases}$$

$$\begin{cases} u(x, 0) = \sum U_n(0) \cdot \sin\left(\frac{n\pi}{l}x\right) = 0 \\ \Rightarrow U_n(0) = 0 \\ u_t(x, 0) = \sum U_n'(0) \cdot \sin\left(\frac{n\pi}{l}x\right) = 0 \\ \Rightarrow U_n'(0) = 0. \end{cases}$$

$$\Rightarrow U_2 = \frac{1}{4a\pi} \cdot \left(\frac{1}{2a\pi} \sin\left(\frac{2a\pi}{l}t\right) - t \cos\left(\frac{2a\pi}{l}t\right) \right)$$

$$U_n = 0$$

$$\Rightarrow u = \frac{1}{4a\pi} \cdot \left(\frac{1}{2a\pi} \sin\left(\frac{2a\pi}{l}t\right) - t \cos\left(\frac{2a\pi}{l}t\right) \right) \cdot \sin\left(\frac{2\pi x}{l}\right).$$

⑤.
$$\begin{cases} u_{tt} = a^2 u_{xx} + f(x, t), & x \in (0, 1), t > 0, \\ u(0, t) = u_1(t), & u(1, t) = u_2(t) \\ u(x, 0) = p(x), & u_t(x, 0) = \psi(x). \end{cases}$$

 N-H-BC, N-H-IC, N-H-IE.

BC非齐次时候 $\rightarrow$ 构造辅助函数 $w(x, t) = \frac{u_2(t) - u_1(t)}{l} x + u_1(t)$
 使BC齐次, 再对辅助函数展开.

i.e. $u = v + w$.

$$\begin{cases} v_{tt} = a^2 v_{xx} + a^2 w_{xx} + f(x, t) - w_{tt} \\ v(0, t) = 0, & v(1, t) = 0 \\ v(x, 0) = p(x) - w(x, 0), & v_t(x, 0) = \psi(x) - w_t(x, 0). \end{cases}$$

e.g.
$$\begin{cases} u_{tt} = a^2 u_{xx} + p, & x \in (0, 1), t > 0, \\ u(0, t) = 0, & u(1, t) = q \\ u(x, 0) = \frac{q}{l} x, & u_t(x, 0) = 0. \end{cases}$$

 N-H-BC, N-H-IC, N-H-IE.

[解]. 取 $w(x, t) = \frac{u_2(t) - u_1(t)}{l} x + u_1(t)$

$$= \frac{q - 0}{l} x + 0 = \frac{q}{l} x = w(x)$$

$$\Rightarrow \begin{cases} v_{tt} = a^2 v_{xx} + a^2 \cdot 0 + p - 0 \\ \quad = a^2 v_{xx} + p. \\ v(0, t) = 0, & v(1, t) = 0. \\ v(x, 0) = \frac{q}{l} x - \frac{q}{l} x = 0 \\ v_t(x, 0) = 0 - 0 = 0 \end{cases}$$

⑥. 高维问题 $\rightarrow$ 先看求解域. $\begin{cases} \text{矩形.} \\ \text{圆扁环} \rightarrow \text{换参.} \end{cases}$ 然后分离变量.
 注: N-H-BC的高维问题基本不能求解.

a.
$$\begin{cases} u_t = a^2 (u_{xx} + u_{yy}), & x \in (0, p), y \in (0, q), t > 0, \\ u(0, y, t) = u(p, y, t) = 0 \\ u(x, 0, t) = u(x, q, t) = 0, \\ u(x, y, 0) = p(x, y). \end{cases}$$

 H-BC, N-H-IC, H-IE.

[解]. ~~u = XYT~~ $u = XYT, \quad XYT' = a^2 X''YT + a^2 XY''T$

$$\frac{X''}{X} + \frac{Y''}{Y} = \frac{1}{a^2} \frac{T'}{T}$$

$$\text{设 } \frac{X''}{X} = -\mu, \frac{Y''}{Y} = -\nu$$

$$\Rightarrow X'' + \mu X = 0, Y'' + \nu Y = 0,$$

$$\frac{1}{a^2} \frac{T'}{T} = -(\mu + \nu), T' + (\mu + \nu)a^2 T = 0$$

$$X(0)Y(q)T(t) = 0 \Rightarrow X(0) = 0, X(p) = 0, Y(0) = 0, Y(q) = 0$$

$$\Rightarrow \begin{cases} X'' + \mu X = 0 \\ X(0) = 0, X(p) = 0 \end{cases} \quad \textcircled{1} \mu = 0, X = Ax + B$$

$$X(0) = B = 0, X(p) = Ap = 0 \Rightarrow X \equiv 0$$

$$\textcircled{2} \mu > 0, X = A \cos(\sqrt{\mu} x) + B \sin(\sqrt{\mu} x)$$

$$X(0) = A = 0, X(p) = 0 \Rightarrow \sqrt{\mu} p = n\pi$$

$$\mu_n = \left(\frac{n\pi}{p}\right)^2, n = 1, \dots$$

$$\Rightarrow X_n = A_n \sin\left(\frac{n\pi}{p} x\right), n = 1, \dots$$

$$\begin{cases} Y'' + \nu Y = 0 \\ Y(0) = 0 = Y(q) \end{cases} \Rightarrow Y_m = B_m \sin\left(\frac{m\pi}{q} y\right), m = 1, \dots$$

$$\begin{cases} T' + (\mu + \nu)a^2 T = 0 \\ T(0) = \varphi(x, y) \end{cases} \Rightarrow T_{mn} = C_{mn} \cdot e^{-\left[\left(\frac{n\pi}{p}\right)^2 + \left(\frac{m\pi}{q}\right)^2\right] a^2 t}$$

$$\Rightarrow U_{mn} = X_n Y_m T_{mn}$$

$$= A_n \cdot B_n \cdot C_{mn} \cdot \sin\left(\frac{n\pi}{p} x\right) \sin\left(\frac{m\pi}{q} y\right) \cdot e^{-\dots}$$

$$= D_{mn} \cdot \sin\left(\frac{n\pi}{p} x\right) \cdot \sin\left(\frac{m\pi}{q} y\right) \cdot e^{-\dots}$$

$$\Rightarrow U = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} U_{mn}$$

$$= \sum_m \sum_n D_{mn} \cdot \sin\left(\frac{n\pi}{p} x\right) \cdot \sin\left(\frac{m\pi}{q} y\right) \cdot e^{-\dots}$$

$$U(x, y, 0) = \sum_m \sum_n D_{mn} \cdot \sin\left(\frac{n\pi}{p} x\right) \cdot \sin\left(\frac{m\pi}{q} y\right) \cdot 1 = \varphi(x, y)$$

$$\Rightarrow D_{mn} = \frac{4}{pq} \int_0^q \int_0^p \varphi(x, y) \sin\left(\frac{n\pi}{p} x\right) \sin\left(\frac{m\pi}{q} y\right) dx dy$$

极坐标 + 特征函数展开法

$$\textcircled{b} \begin{cases} u_{xx} + u_{yy} = 12(x^2 - y^2), \sqrt{x^2 + y^2} \in (a, b) \\ u|_{\sqrt{x^2 + y^2} = a} = u|_{\sqrt{x^2 + y^2} = b} = 0 \end{cases}$$

$$[\text{极坐标}]. \begin{cases} \rho^2 = x^2 + y^2 \\ \theta = \arctan \frac{y}{x} \end{cases} \quad \begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta \end{cases}$$

$$u = \rho \theta$$

$$\rightarrow \rho'' \theta + \frac{1}{\rho} \rho' \theta + \frac{1}{\rho^2} \rho \theta'' = 0$$

极坐标下的拉普拉斯形式

$$\frac{p^2 p'' + p p'}{p} = -\frac{\theta''}{\theta} = \lambda.$$

$$\begin{cases} \theta'' + \lambda \theta = 0 \\ \theta(0) = \theta(\theta + 2\pi) \end{cases} \Rightarrow \lambda_n = n^2, n=0, \dots$$

$$\theta_n = A_n \cos(n\theta) + B_n \sin(n\theta), n=0, \dots$$

$$p^2 p'' + p p' - \lambda p = 0.$$

$$\Rightarrow u(p, \theta) = \sum_{n=0}^{\infty} A_n(p) \cdot \cos(n\theta) + B_n(p) \sin(n\theta)$$

$$u_{pp} + \frac{1}{p} u_p + \frac{1}{p^2} u_{\theta\theta} = 12 p^2 (\cos^2 \theta - \sin^2 \theta) = 12 p^2 \cos 2\theta.$$

$$u(a, \theta) = 0 = u(b, \theta) \Rightarrow p(a) = p(b) = 0.$$

$$\sum_{n=0}^{\infty} \left[A_n'' \cdot \cos(n\theta) + B_n'' \cdot \sin(n\theta) + \frac{1}{p} A_n' \cdot \cos(n\theta) + \frac{1}{p} B_n' \cdot \sin(n\theta) + \frac{1}{p^2} A_n \cdot (-n^2) \cdot \cos(n\theta) + \frac{1}{p^2} B_n \cdot (-n^2) \cdot \sin(n\theta) \right] = 12 p^2 \cos(2\theta)$$

$$n=2: \begin{cases} A_2'' + \frac{1}{p} A_2' - \frac{4}{p^2} A_2 = 12 p^2 \\ B_2'' + \frac{1}{p} B_2' - \frac{4}{p^2} B_2 = 0 \end{cases} \Rightarrow B_2 = 0$$

$$n \neq 2: \begin{cases} A_n'' + \frac{1}{p} A_n' - \frac{n^2}{p^2} A_n = 0 \\ B_n'' + \frac{1}{p} B_n' - \frac{n^2}{p^2} B_n = 0 \end{cases} \Rightarrow A_n = 0, B_n = 0.$$

$$p^2 A_2'' + p A_2' - 4 A_2 = 12 p^4$$

$$p^2 A_2'' + A_2' - 4 A_2 = 0.$$

$$\text{Euler. } t = \ln p \Rightarrow A_{2+t} - 4 A_2 = 0$$

$$\Rightarrow A_{2h} = C_2 \cdot e^{27h} + D_2 \cdot e^{-27h}$$

$$= C_2 \cdot p^2 + D_2 \cdot p^{-2}$$

$$\text{设 } A_{2p} = E \cdot p^4.$$

$$A_{2p}' = 4E \cdot p^3, A_{2p}'' = 12E \cdot p^2.$$

$$p^2 \cdot 12E p^2 + p \cdot 4E \cdot p^3 - 4E \cdot p^4 = 12 p^4$$

$$12E \cdot p^4 = 12 p^4 \Rightarrow E = 1$$

$$\Rightarrow A_2 = A_{2h} + A_{2p} = C_2 \cdot p^2 + D_2 \cdot p^{-2} + p^4$$

$$\Rightarrow u = \sum_{n=0}^{\infty} (C_2 \cdot p^2 + D_2 \cdot p^{-2} + p^4) \cdot \cos(n\theta).$$

$$\begin{cases} p(a) = C \cdot a^2 + D \cdot a^{-2} + a^4 = 0 \\ p(b) = C \cdot b^2 + D \cdot b^{-2} + b^4 = 0 \end{cases} \Rightarrow \begin{cases} C_2 = -\frac{a^4 b - b^4}{a^4 - b^4} \\ D_2 = -\frac{a^4 b^4}{a^2 + b^2} \end{cases}$$

5. Green 函数.

无穷级数解在实际中需做截断.

变换法所得广义积分解实用时需变换

→ Green 函数法给出定积分解.

Def. u 调和 (harmonic) $\Leftrightarrow \Delta u = u_{xx} + u_{yy} + u_{zz} = 0$. i.e. Lap/调和方程 $= 0$.

(1). 取 V s.t. $\begin{cases} P = u \cdot V_x \\ Q = u \cdot V_y \\ R = u \cdot V_z \end{cases} \Rightarrow \iiint_{\Omega} (u \Delta V) dx dy dz + \iiint_{\Omega} (\nabla u \cdot \nabla V) dx dy dz = \iint_{\partial \Omega} u \left(\frac{\partial V}{\partial n} \right) dS \dots \textcircled{1}$ → 曲面积分

I. Green := $\iiint_{\Omega} (u \Delta V) dx dy dz + \iiint_{\Omega} (\nabla u \cdot \nabla V) dx dy dz = \iint_{\partial \Omega} u \left(\frac{\partial V}{\partial n} \right) dS$.

(2). 由 I. Green 有 $\iiint_{\Omega} (V \Delta u) dx dy dz + \iiint_{\Omega} (\nabla V \cdot \nabla u) dx dy dz = \iint_{\partial \Omega} V \left(\frac{\partial u}{\partial n} \right) dS \dots \textcircled{2}$

$\textcircled{1} - \textcircled{2} \Rightarrow \iiint_{\Omega} (u \Delta V - V \Delta u) dx dy dz = \iint_{\partial \Omega} \left(u \frac{\partial V}{\partial n} - V \frac{\partial u}{\partial n} \right) dS$.

即为 I. Green.

Theo 1. u 在 Ω 调和 $\Rightarrow \iint_{\partial \Omega} \frac{\partial u}{\partial n} dS = 0$. i.e. 在曲面/曲线上积分为 0.

e.g. $\begin{cases} \Delta u = 0 \\ u_x(0, y) = f_1(y), u_x(a, y) = f_2(y) \\ u_y(x, 0) = g_1(x), u_y(x, b) = g_2(x) \end{cases} \quad x \in (0, a), y \in (0, b).$

u 调和且 $BC(N) \Rightarrow$ 必有 $\int_0^a (g_1 - g_2) dx + \int_0^b (f_1 - f_2) dy = \iint \frac{\partial u}{\partial n} dS = 0$.

6. Lap 基本解.

Def. Lap 方程的基本解 := 在某种对称性下考虑的对称解. 可作厚点的点电荷.

2D. $\begin{cases} \Delta u = u_{xx} + u_{yy} = 0 \\ u_{pp} + \frac{1}{p} \cdot u_p + \frac{1}{p^2} u_{\theta\theta} = 0 \end{cases} \Rightarrow V_0(r) = -\frac{1}{2\pi} \ln \frac{1}{r}$.

3D. $\begin{cases} \Delta u = u_{xx} + u_{yy} + u_{zz} = 0 \\ \frac{1}{p^2} \cdot \frac{\partial}{\partial p} \cdot (r^2 u_p) + \frac{1}{r^2 \sin \theta} \cdot \frac{\partial}{\partial \theta} (\sin \theta \cdot u_{\theta}) + \frac{1}{r^2 \sin^2 \theta} \cdot u_{pp} = 0 \end{cases} \Rightarrow V_0(r) = -\frac{1}{4\pi r}$

r_{MM_0} : M 到 M_0 的欧氏距离.

$\Rightarrow V(r) = -\frac{1}{4\pi \sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2}}$
 $= -\frac{1}{4\pi r_{MM_0}}$

Theo 2. $\forall M_0(x_0, y_0, z_0) \in \Omega$,

$$u(M_0) = \frac{1}{4\pi} \iint_{\partial\Omega} \left(\frac{1}{r_{MM_0}} \cdot \frac{\partial u}{\partial n} - u \cdot \frac{\partial}{\partial n} \left(\frac{1}{r_{MM_0}} \right) \right) dS \\ - \frac{1}{4\pi} \iiint_{\Omega} \frac{\Delta u}{r_{MM_0}} dx dy dz.$$

Le m 1. $\Delta u = 0, \forall M_0(x_0, y_0, z_0) \in \Omega$,

$$u(M_0) = \frac{1}{4\pi} \iint_{\partial\Omega} \left(\frac{1}{r_{MM_0}} \cdot \frac{\partial u}{\partial n} - u \cdot \frac{\partial}{\partial n} \left(\frac{1}{r_{MM_0}} \right) \right) dS.$$

Le m 2. $\Delta u = f(x, y, z), \forall M_0(x_0, y_0, z_0) \in \Omega$,

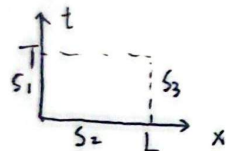
$$u(M_0) = \frac{1}{4\pi} \iint_{\partial\Omega} \left(\frac{1}{r_{MM_0}} \cdot \frac{\partial u}{\partial n} - u \cdot \frac{\partial}{\partial n} \left(\frac{1}{r_{MM_0}} \right) \right) dS \\ - \frac{1}{4\pi} \iiint_{\Omega} \frac{f(x, y, z)}{r_{MM_0}} dx dy dz.$$

7. 球面平均值定理.

$$\text{Theo 3. } \Delta u = 0 \Rightarrow u(M_0) = \frac{1}{4\pi a^2} \iint_{\partial B_a(M_0)} u dS.$$

8. 极值原理: 调和 u 的极大极小值只能在边界上取到.

对热方程的极值原理: 限制 t 上限 的矩形域最值在 U 形边界上取到.



$$S = S_1 + S_2 + S_3.$$

$$Q = \{(x, t) \mid 0 < x < L, 0 < t \leq T\}.$$

$$\Rightarrow \begin{cases} \max_Q u(x, t) = \max_S u(x, t) \\ \min_Q u(x, t) = \min_S u(x, t) \end{cases}$$

Ex 1. $\begin{cases} \Delta u = f(x, y, z) \\ u|_{\partial\Omega} = g(x, y, z). \end{cases}$

[证]. ① 唯一性. 设有两解 u_1, u_2 .

$$u = u_1 - u_2 \Rightarrow \begin{cases} \Delta u = 0 \\ u|_{\partial\Omega} = 0. \end{cases}$$

由极值原理, $0 = u_{\max} = u_{\min} \Rightarrow u = 0$.

Δu 已调和即默认满足基本光滑性假设.

②. 关于 B 连续.

$$\begin{cases} \Delta u_1 = f(x, y, z) \\ u_1|_{\partial\Omega} = g_1(x, y, z) \end{cases} \quad \begin{cases} \Delta u_2 = f(x, y, z) \\ u_2|_{\partial\Omega} = g_2(x, y, z) \end{cases}$$

$\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) \text{ s.t. } \forall |g_1 - g_2| < \delta \text{ 有}$

$$|u| = |u_1 - u_2| \leq |g_1 - g_2| < \delta < \varepsilon. \quad \square$$

由极值原理.

Ex2. 补区域 $\Omega^c = \mathbb{R}^3 \setminus \Omega$.
$$\begin{cases} \Delta u = g(x, y, z), & (x, y, z) \in \Omega^c \\ u(x, y, z) = f(x, y, z) \\ \lim_{r \rightarrow \infty} u(x, y, z) = 0. \end{cases}$$

证解的存在与唯一性。
能定。

[解]. 思路与上题一致。

极值原理在无界区域也成立 (极值在 内区域的边界 或 外边界 取到)

$u=0$.

此题 $|u| < \infty$

Ex3.
$$\begin{cases} u_t = c^2 u_{xx} + h(x, t), & x \in (0, L), t > 0 \\ u(x, 0) = f(x) \\ u(0, t) = p(t), u(L, t) = q(t). \end{cases}$$

[解]. 思路与上题一致。

Ex4. 外边界问题.
$$\begin{cases} u_t = a^2 u_{xx}, & x > 0, t > 0 \\ u(x, 0) = 0 \\ u(0, t) = N, u(+\infty, t) = 0. \end{cases}$$

N-H-B.C. H-I.C. H-E.C.

[解]. $\mathcal{L}[u(x, t)] = U(x, s)$

$$\mathcal{L}[u_t] = s \cdot U - u(x, 0) = \mathcal{L}[u_{xx}] \Rightarrow s \cdot U = a^2 u_{xx}$$

$$\mathcal{L}[u(0, t)] = U(0, s) = N$$

$$\mathcal{L}[u(+\infty, t)] = U(+\infty, s) = 0$$

$$\begin{cases} U_{xx} - \frac{s}{a^2} U = 0 \\ U(0, s) = N, U(+\infty, s) = 0. \end{cases}$$

$$\Rightarrow U = A \cdot e^{\frac{1}{a}\sqrt{s}x} + B \cdot e^{-\frac{1}{a}\sqrt{s}x}$$

$$U(0, s) = A + B = N, U(+\infty, s) = 0 \Rightarrow A = 0$$

$$\Rightarrow U = N \cdot e^{-\frac{1}{a}\sqrt{s}x}$$

$$\Rightarrow u(x, t) = \mathcal{L}^{-1}[U(x, s)] = \mathcal{L}^{-1}[N \cdot e^{-\frac{1}{a}\sqrt{s}x}]$$

□

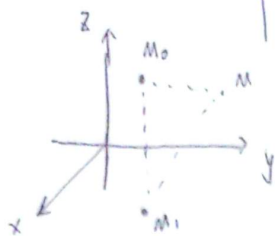
9. 解 BC(D) 的 Lap $\rightarrow$ 针对有限区域构造 Green 函数, 得到有限积分下的解.

$$\begin{cases} \Delta u = 0, & x \in \Omega \\ u|_{\partial\Omega} = f. \end{cases} \Rightarrow \text{只需求出 } v \text{ s.t. } \begin{cases} \Delta v = 0 & \text{in } \Omega \\ v|_{\partial\Omega} = -\frac{1}{4\pi r_{MM_0}} \end{cases}$$

$$\text{则 Green 函数 } G(M, M_0) = v + \frac{1}{4\pi r_{MM_0}}, \begin{cases} \Delta G = 0 & \text{in } \Omega \setminus \{M_0\} \\ G|_{\partial\Omega} = 0 \end{cases}$$

$$\text{解为 } u(M_0) = - \int_{\partial\Omega} f \cdot \frac{\partial G}{\partial n} \cdot dS$$

(1). 3维电势. $\begin{cases} \Delta u = 0, z > 0, x, y \in \mathbb{R} \\ u(x, y, 0) = f(x, y) \end{cases}$



取 M_0 关于半球区域 对称点/像点 为 M_1 , 则

$$G = -\frac{1}{4\pi r_{MM_0}} + \frac{q}{4\pi r_{MM_1}} \quad (\text{由物理含义可知, 视 } M_0 \text{ 为正电荷, } M_1 \text{ 为负电荷, 则 } G \text{ 仍得证在 } \Omega \setminus \{M_0\} \text{ 调和}).$$

(2). 2维. $\begin{cases} \Delta u = 0, x \in \mathbb{R}, y > 0 \\ u(x, 0) = f(x) \end{cases}$

$$G = -\frac{1}{2\pi} \ln \frac{1}{r_{MM_0}} - \frac{q}{2\pi} \ln \frac{1}{r_{MM_1}}$$

$$\text{s.t. } \begin{cases} \Delta G = 0 & \text{in } D \setminus \{M_0\} \\ G|_{\partial D} = 0 \end{cases}$$

(3). 球. $\begin{cases} \Delta u = 0, (x, y, z) \in \Omega \\ u|_{\partial\Omega} = f(x, y, z) \end{cases}$

$$G = \frac{1}{4\pi} \left(\frac{1}{r_{MM_0}} - \frac{R}{\rho} \frac{1}{r_{MM_1}} \right),$$

$$q = \frac{r_{MM_1}}{r_{M_0M}} = \frac{R}{\rho}.$$

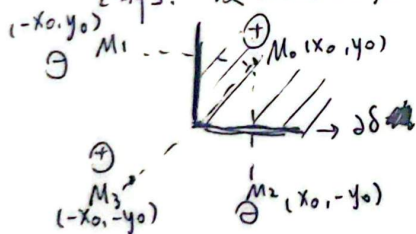
(4). 圆. $\begin{cases} \Delta u = 0, (x, y) \in \Omega \\ u|_{\partial\Omega} = f(x, y) \end{cases}$

$$G = \frac{1}{2\pi} \left(\ln \frac{1}{r_{MM_0}} - \ln \frac{R}{r_{OM_0}} \cdot \frac{1}{r_{MM_1}} \right).$$

Ex1. $\begin{cases} u_{xx} + u_{yy} = 0, x, y > 0 \\ u(0, y) = 0 \\ u(x, 0) = f(x) \end{cases}$

求 G 及 u .

[解]. 设 $M_0(x_0, y_0) \in \Omega$.



$$\Rightarrow G = -\frac{1}{2\pi} \left(\ln \frac{1}{r_{MM_0}} + \ln \frac{1}{r_{MM_1}} - \ln \frac{1}{r_{MM_2}} + \ln \frac{1}{r_{MM_3}} \right)$$

$$\Rightarrow u(M_0) = - \int_{y=0}^{\infty} u(x, 0) \frac{\partial G}{\partial n} dS + \int_{x=0}^{\infty} u(0, y) \frac{\partial G}{\partial n} dS$$

$$= \int_0^{\infty} f(x) \cdot \frac{\partial G}{\partial y} \Big|_{y=0} dx$$

$$= \int_0^{\infty} f(x) \cdot \frac{y_0}{\pi} \left(\frac{1}{(x-x_0)^2 + y_0^2} - \frac{1}{(x+x_0)^2 + y_0^2} \right) dx$$

$$\Rightarrow u(x, y) = \frac{y}{\pi} \int_0^{\infty} f(x) \cdot \left(\frac{1}{(x-x_0)^2 + y^2} - \frac{1}{(x+x_0)^2 + y^2} \right) dx.$$

像点可以不只一个, 只需保证边界上电位=0

$$\text{Ex 2. } \begin{cases} \Delta u(r, \theta, \rho) = 0 & r \in (r_1, r_2) \\ u|_{r=r_1} = V_0, u|_{r=r_2} = 0. \end{cases}$$

[解]. 问题球对称且与 θ, φ 无关 $\Rightarrow \Delta u = \frac{1}{\rho^2} \cdot \frac{\partial}{\partial \rho} \cdot (\rho^2 u_\rho) = 0$

$$\Rightarrow \frac{\partial}{\partial \rho} (\rho^2 u_\rho) = 0$$

$$\rho^2 u_\rho = C$$

$$u_\rho = \frac{1}{\rho^2} C$$

$$\Rightarrow u(\rho) = \frac{A}{\rho} + B$$

$$BC \Rightarrow u = u(r) = V_0 \frac{\frac{r_2}{r} - 1}{\frac{r_2}{r_1} - 1}.$$

///

Chapter 2. 热方程.

1. 分离变量法 $\rightarrow$ H-BC, N-H-IC, H-E 热.

①. H-BC, N-H-IC, H-E 热.
(R)

$$\begin{cases} U_t = a^2 U_{xx} \\ U(x, 0) = \varphi(x) \\ U(0, t) = 0, U_x(l, t) + hU(l, t) = 0. \end{cases}$$

[解] $U(x, t) = XT, \quad XT' = a^2 X''T, \quad \frac{X''}{X} = \frac{T'}{a^2 T} = -\lambda.$

$$\begin{cases} X'' + \lambda X = 0 \\ T' + a^2 \lambda T = 0 \end{cases} \quad \begin{cases} X(0)T(t) = 0 \Rightarrow X(0) = 0 \\ X'(l)T(t) + hX(l)T(t) = 0 \Rightarrow X'(l) + hX(l) = 0. \end{cases}$$

$$\Rightarrow \begin{cases} X'' + \lambda X = 0 \\ X'(l) + hX(l) = 0, X(0) = 0. \end{cases}$$

①. $\lambda = 0, X = Ax + B, X(0) = B = 0, A + hAl + hB = 0 = A(1 + hl) \Rightarrow A = 0 \Rightarrow X \equiv 0.$

②. $\lambda < 0, X = A \cdot e^{\sqrt{\lambda}x} + B \cdot e^{-\sqrt{\lambda}x}.$

$$\begin{aligned} X(0) = A + B = 0, X'(l) + hX(l) &= \sqrt{\lambda} \cdot A \cdot e^{\sqrt{\lambda}l} - \sqrt{\lambda} \cdot B \cdot e^{-\sqrt{\lambda}l} + h \cdot A \cdot e^{\sqrt{\lambda}l} + h \cdot B \cdot e^{-\sqrt{\lambda}l} \\ &= A(h + \sqrt{\lambda}) \cdot e^{\sqrt{\lambda}l} + B(h - \sqrt{\lambda}) \cdot e^{-\sqrt{\lambda}l} \end{aligned}$$

$$\Rightarrow B = A = 0 \Rightarrow X \equiv 0.$$

③. $\lambda > 0, X = A \cdot \cos \sqrt{\lambda}x + B \sin \sqrt{\lambda}x.$

$$\begin{aligned} X(0) = A = 0, X'(l) + hX(l) &= \sqrt{\lambda} B \cdot \cos(\sqrt{\lambda}l) + hB \sin(\sqrt{\lambda}l) \\ &= B(\sqrt{\lambda} \cos(\sqrt{\lambda}l) + h \sin(\sqrt{\lambda}l)) = 0 \end{aligned}$$

$$B \neq 0 \Rightarrow h \sin(\sqrt{\lambda}l) = -\sqrt{\lambda} \cos(\sqrt{\lambda}l).$$

$$\tan(\sqrt{\lambda}l) = -\frac{\sqrt{\lambda}}{h}.$$

$$\text{取 } \gamma = \sqrt{\lambda}l, \alpha = \frac{1}{hl} \Rightarrow \tan \gamma = \tan(\sqrt{\lambda}l) = -\frac{\sqrt{\lambda}}{h} = \left(-\frac{1}{hl}\right) \cdot (\sqrt{\lambda}l)$$

$$\text{设 } 0 < \gamma_1 < \gamma_2 < \dots < \gamma_n \Rightarrow \lambda_n = \frac{\gamma_n^2}{l^2}, n = 1, 2, \dots$$

$$\Rightarrow X_n = B_n \sin(\sqrt{\lambda_n}x) = B_n \sin\left(\frac{\gamma_n}{l}x\right), n = 1, 2, \dots$$

$$T' + a^2 \lambda T = 0$$

$$\int \frac{1}{T} dT = -\int a^2 \lambda dt$$

$$T = e^{-a^2 \lambda t} \Rightarrow T_n = e^{-a^2 \lambda_n t} = e^{-a^2 \left(\frac{\gamma_n^2}{l^2}\right)t}.$$

$$\Rightarrow U_n = X_n T_n = B_n \sin(\sqrt{\lambda_n}x) \cdot e^{-a^2 \lambda_n t}$$

$$\Rightarrow U = \sum_{n=1}^{\infty} U_n = \sum A_n \cdot \sin(\sqrt{\lambda_n}x) \cdot e^{-a^2 \lambda_n t}$$

$$U(x, 0) = \sum_{n=1}^{\infty} A_n \cdot \sin(\sqrt{\lambda_n}x) = \varphi(x)$$

$$A_n \int_0^l \sin^2(\sqrt{\lambda_n}x) dx = \int_0^l \varphi(x) \sin(\sqrt{\lambda_n}x) dx$$

$$\Rightarrow A_n = \frac{\int_0^l \varphi(x) \sin(\sqrt{\lambda_n}x) dx}{\int_0^l \sin^2(\sqrt{\lambda_n}x) dx} = \frac{\int_0^l \varphi(x) \cdot \sin(\sqrt{\lambda_n}x) dx}{\frac{l}{2} + \frac{h}{2\lambda_n} \sin^2(\sqrt{\lambda_n}l)}$$

此题 BC 为 (D) 和 (R) 而非简单 (D), (R)
所以级数收敛

②. H-BC. NH-IC. H-E 型.

(D)

$$\begin{cases} u_t = a^2 u_{xx} \\ u(0, t) = 0, u(l, t) = 0 \\ u(x, 0) = \varphi(x) \end{cases}$$

[解] $u = XT, \quad XT' = a^2 X''T, \quad \frac{X''}{X} = \frac{T'}{a^2 T} = -\lambda.$

$$\begin{cases} X'' + \lambda X = 0 \\ T' + \lambda a^2 T = 0 \end{cases} \quad \begin{cases} X(0)T(t) = 0 \Rightarrow X(0) = 0 \\ X(l)T(t) = 0 \Rightarrow X(l) = 0 \end{cases}$$

$$\begin{cases} X'' + \lambda X = 0 \\ X(0) = X(l) = 0 \end{cases} \quad \text{①. } \lambda = 0, X = Ax + B, B = Al + B = 0 \Rightarrow X \equiv 0.$$

$$\text{②. } \lambda < 0, X = A \cdot e^{\sqrt{-\lambda}x} + B \cdot e^{-\sqrt{-\lambda}x}$$

$$X(0) = A + B = X(l) = A \cdot e^{-\sqrt{-\lambda}l} + B \cdot e^{-\sqrt{-\lambda}l} = 0 \Rightarrow X \equiv 0.$$

$$\text{③. } \lambda > 0, X = A \cos(\sqrt{\lambda}x) + B \sin(\sqrt{\lambda}x)$$

$$X(0) = A = 0, X(l) = B \sin(\sqrt{\lambda}l) = 0 \Rightarrow \sqrt{\lambda}l = n\pi, n=1, \dots$$

$$\Rightarrow \lambda_n = \left(\frac{n\pi}{l}\right)^2, n=1, \dots$$

$$\Rightarrow X_n = B_n \sin\left(\frac{n\pi x}{l}\right), n=1, \dots$$

$$T' + \lambda a^2 T = 0.$$

$$\frac{1}{T} dT = -\lambda a^2 dt$$

$$T = e^{-\lambda a^2 t} \Rightarrow T_n = e^{-\lambda_n a^2 t} = e^{-\left(\frac{n\pi}{l}\right)^2 a^2 t}.$$

$$\Rightarrow u_n = A_n \sin\left(\frac{n\pi x}{l}\right) \cdot e^{-\left(\frac{n\pi}{l}\right)^2 a^2 t}.$$

$$\Rightarrow u = \sum_{n=1}^{\infty} u_n = \sum A_n \sin\left(\frac{n\pi x}{l}\right) \cdot e^{-\left(\frac{n\pi}{l}\right)^2 a^2 t}.$$

$$u(x, 0) = \sum A_n \sin\left(\frac{n\pi x}{l}\right) = \varphi(x)$$

$$A_n \int_0^l \sin^2\left(\frac{n\pi x}{l}\right) dx = \int_0^l \varphi(x) \sin\left(\frac{n\pi x}{l}\right) dx.$$

$$A_n = \frac{2}{l} \int_0^l \varphi(x) \sin\left(\frac{n\pi x}{l}\right) dx.$$

▣

③ H-BC. NH-IC. H-E 型.

(N)

$$\begin{cases} u_t = a^2 u_{xx} \\ u_x(0, t) = 0, u_x(l, t) = 0 \\ u(x, 0) = \varphi(x) \end{cases}$$

[解] $\begin{cases} X'' + \lambda X = 0 \\ T' + \lambda a^2 T = 0 \end{cases} \quad \begin{cases} X'(0)T(t) = 0 \Rightarrow X'(0) = 0 \\ X'(l)T(t) = 0 \Rightarrow X'(l) = 0 \end{cases}$

$$1. X'' + \lambda X = 0 \quad \text{①. } \lambda = 0, X = Ax + B, \\ \begin{cases} X'(l) = X'(0) = 0, & X'(0) = A = 0 \Rightarrow X = B \Rightarrow \lambda_0 = 0, X_0 = B_0. \end{cases}$$

$$\text{②. } \lambda < 0, X = A \cdot e^{\sqrt{-\lambda}x} + B \cdot e^{-\sqrt{-\lambda}x}.$$

$$X' = \sqrt{-\lambda} \cdot A \cdot e^{\sqrt{-\lambda}x} - \sqrt{-\lambda} \cdot B \cdot e^{-\sqrt{-\lambda}x} \quad X'(0) = \sqrt{-\lambda} \cdot A - \sqrt{-\lambda} \cdot B = 0, \\ X'(l) = \sqrt{-\lambda} \cdot A \cdot e^{\sqrt{-\lambda}l} - \sqrt{-\lambda} \cdot B \cdot e^{-\sqrt{-\lambda}l} = 0 \Rightarrow X \equiv 0.$$

$$\text{③. } \lambda > 0, X = A \cos(\sqrt{\lambda}x) + B \sin(\sqrt{\lambda}x).$$

$$X' = -A \cdot \sqrt{\lambda} \sin(\sqrt{\lambda}x) + B \cdot \sqrt{\lambda} \cos(\sqrt{\lambda}x).$$

$$X'(0) = B\sqrt{\lambda} = 0 \Rightarrow B = 0, X'(l) = -A \cdot \sqrt{\lambda} \sin(\sqrt{\lambda}l) = 0.$$

$$\Rightarrow \sqrt{\lambda}l = n\pi, \Rightarrow \lambda_n = \left(\frac{n\pi}{l}\right)^2, n = 1, \dots, \lambda_0 = 0$$

$$\Rightarrow X_n = A_n \cos\left(\frac{n\pi}{l}x\right), n = 1, \dots, X_0 = A_0.$$

$$T_n = e^{-\lambda_n a^2 t} = e^{-\left(\frac{n\pi}{l}\right)^2 a^2 t}, n = 1, \dots, T_0 = e^0 = 1.$$

$$\Rightarrow u = \sum_{n=0}^{\infty} X_n T_n = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi}{l}x\right) \cdot e^{-\left(\frac{n\pi}{l}\right)^2 a^2 t}.$$

$$u(x, 0) = A_0 + \sum_{n=1}^{\infty} A_n \cdot \cos\left(\frac{n\pi}{l}x\right) = \varphi(x). \rightarrow \varphi(x) \text{ 的余弦级数展开.}$$

$$\Rightarrow A_0 = \frac{1}{l} \int_0^l \varphi(x) dx, A_n = \frac{2}{l} \int_0^l \varphi(x) \cos\left(\frac{n\pi}{l}x\right) dx, n = 1, \dots$$

$$\Rightarrow u = \frac{1}{l} \int_0^l \varphi(x) dx + \sum_{n=1}^{\infty} \left(\frac{2}{l} \int_0^l \varphi(x) \cos\left(\frac{n\pi}{l}x\right) dx \right) \cdot \cos\left(\frac{n\pi}{l}x\right) \cdot e^{-\left(\frac{n\pi}{l}\right)^2 a^2 t}$$

□

2.

①. Fourier 变换. $(x \in \mathbb{R}).$

$$F[f] = \int_{-\infty}^{\infty} f(x) \cdot e^{-ix\gamma} dx = \tilde{f}(\gamma).$$

$$F^{-1}[\tilde{f}] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(\gamma) \cdot e^{ix\gamma} d\gamma = f(x).$$

(i) 线性. $F[\alpha f + \beta g] = \alpha F[f] + \beta F[g].$

(ii). 可微. (1). $F[f'] = i\gamma F[f].$

(2). $F[f^{(n)}] = (i\gamma)^n F[f].$

(iii). $F[f(x-c)] = \int_{-\infty}^{\infty} f(x-c) e^{-ix\gamma} dx$
 $= \int_{-\infty}^{\infty} f(t) \cdot e^{-i(t+c)\gamma} dt$
 $= e^{-ic\gamma} \cdot F[f(x)].$

$x \in \mathbb{R}^n$ 时有 $F[f] = \int_{\mathbb{R}^n} f(r) \cdot e^{-ir\gamma} dr = \tilde{f}(\gamma)$

$$F^{-1}[\tilde{f}] = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \tilde{f}(\gamma) e^{ir\gamma} d\gamma = f(r).$$

②. 卷积变换. $f * g(x) = \int_{-\infty}^{\infty} f(x-t)g(t)dt$,

(i). 交换. $f * g(x) = g * f(x)$.

(ii). 结合. $(f * g) * h = g * (f * h)$

(iii). 分配. $f * (g+h)(x) = f * g(x) + f * h(x)$.

(iv). 卷积定理.

(1). $F[f * g(x)] = F[f(x)] F[g(x)]$.

$$F^{-1}(F[f(x)] F[g(x)]) = F^{-1}[F[f(x)] * F^{-1}[F[g(x)]]] \\ = (f * g)(x).$$

(2). $F[f(x)g(x)] = \frac{1}{2\pi} F[f(x)] * F[g(x)]$.

$$F^{-1}(F[f(x)] * F[g(x)]) = 2\pi f(x)g(x).$$

③.

(1). 无限 N-H-1C, H-E 热.

$$\begin{cases} ut - a^2 u_{xx} = 0, (x, t) \in \mathbb{R} \times \mathbb{R}^+ \\ u(x, 0) = \varphi(x). \end{cases}$$

[解]. $F[ut] - a^2 F[u_{xx}] = 0$.

$$\frac{dF[u]}{dt} + a^2 \zeta^2 F[u] = 0.$$

$$F[u]|_{t=0} = F[\varphi](\zeta).$$

$$\frac{1}{F[u]} dF[u] = -a^2 \zeta^2 dt$$

$$\ln |F[u]| = -a^2 \zeta^2 t + C$$

$$F[u] = C \cdot e^{-a^2 \zeta^2 t}.$$

$$F[u]|_{t=0} = C = F[\varphi](\zeta)$$

$$\Rightarrow F[u] = F[\varphi](\zeta) \cdot e^{-a^2 \zeta^2 t}$$

或

$$\text{设 } u(x, t) = \phi(x) * h(x, t) \\ = \int_{-\infty}^{\infty} \phi(s) h(x-s, t) ds.$$

$$\text{由 } F[u] = F[\varphi](\zeta) \cdot e^{-a^2 \zeta^2 t} \\ = F[\varphi] \cdot F[h]$$

$$\Rightarrow h(x, t) = F^{-1}[e^{-a^2 \zeta^2 t}] \\ = \frac{1}{2a\sqrt{\pi t}} \cdot e^{-\frac{x^2}{4a^2 t}}$$

$$\Rightarrow u(x, t) = \frac{1}{2a\sqrt{\pi t}} \cdot \int_{-\infty}^{\infty} \phi(s) e^{-\frac{(x-s)^2}{4a^2 t}} ds.$$

